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BA 12.0023
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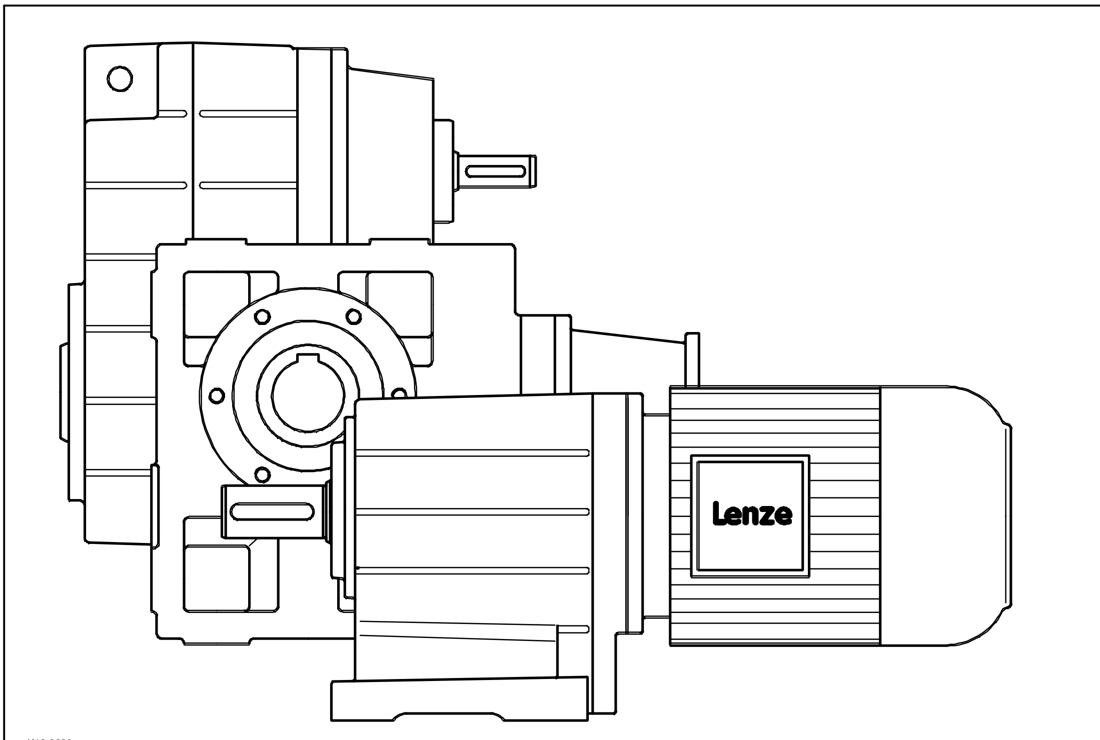
Lenze

Operating Instructions



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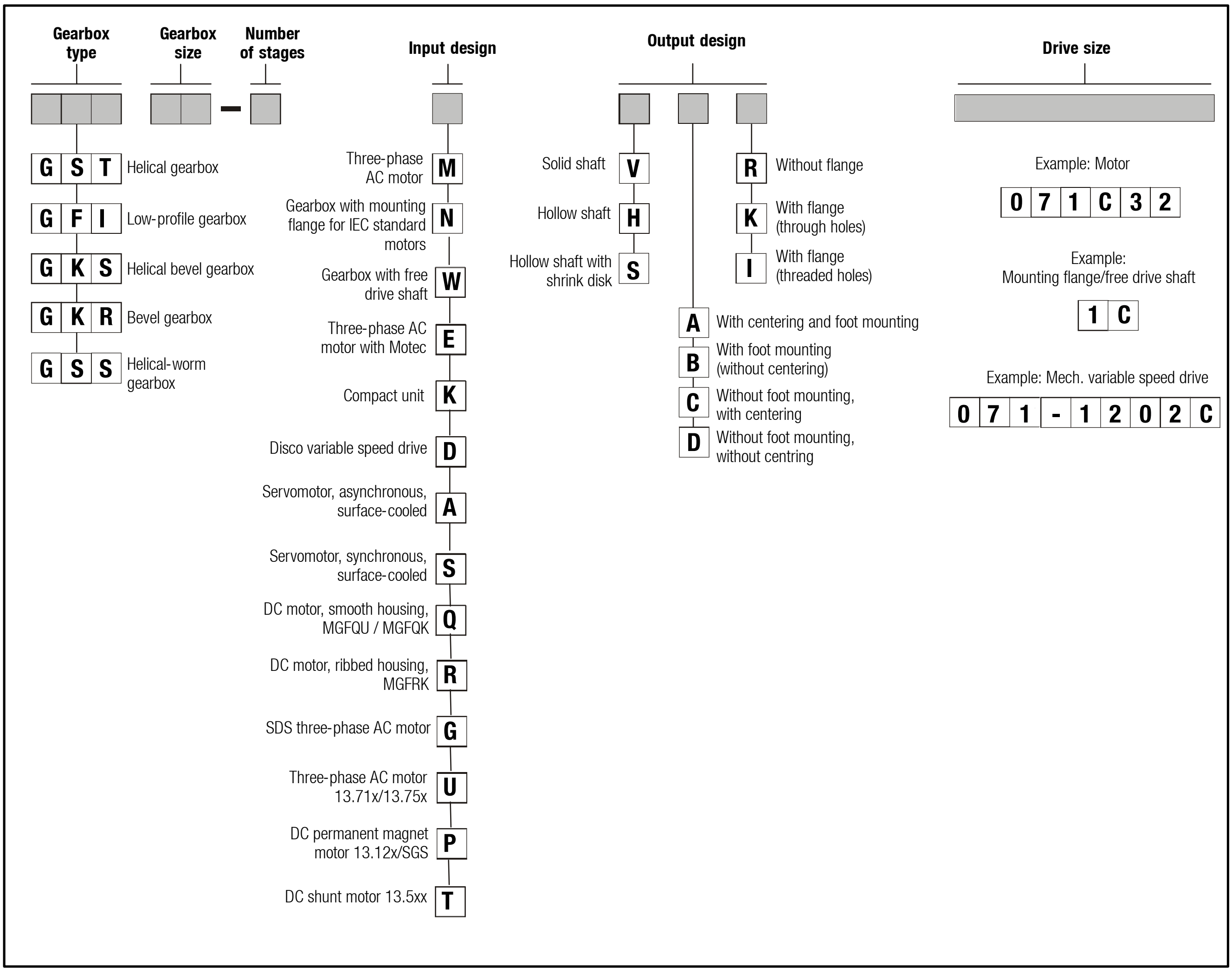
G-motion

Gearbox G☐☐

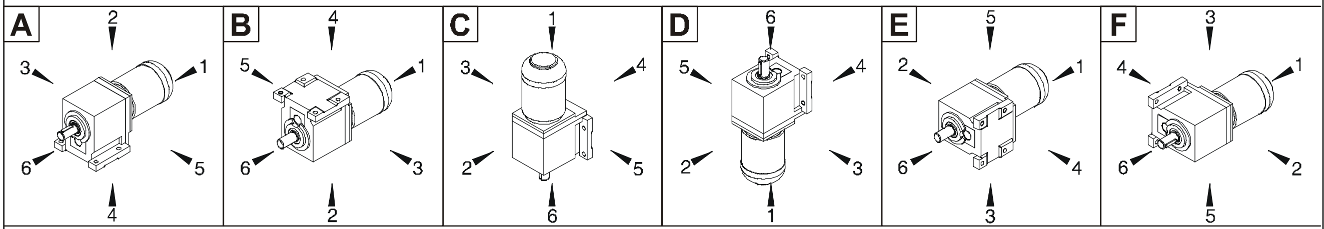
Product key



BA 12.0023
Author: Lenze Drive Systems GmbH
1st edition: 10/03

Mounting position (A-F) and position of system modules (1-6)

GST Terminal box: 2, 3, 4, 5
Without terminal box: 0

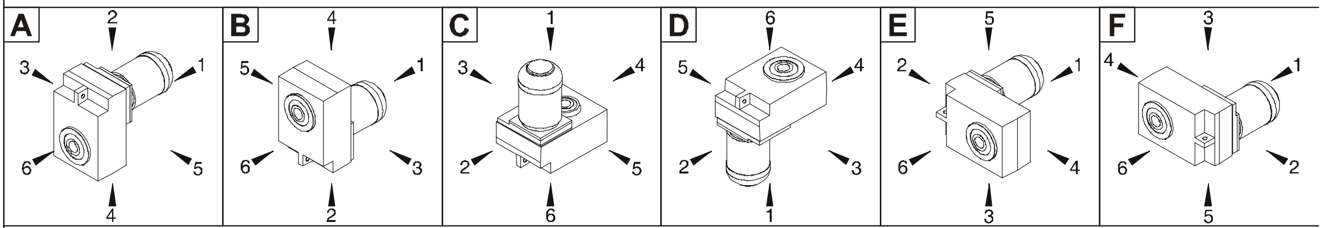


GFL

Solid shaft: 6
Hollow shaft: 0
Hollow shaft with shrink disk: 1, 6

Foot: 3, 4
Without foot: 0

Terminal box: 2, 3, 4, 5
Without terminal box: 0

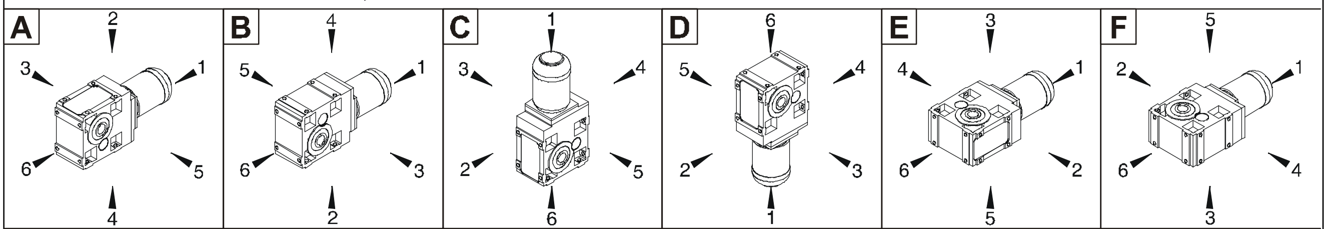


GKS/ GSS

Solid shaft: 3, 5, 3+5
Hollow shaft: 0
Hollow shaft with shrink disk: 3, 5

Flange: 3, 5, 3+5
Without flange: 0

Terminal box: 2, 3, 4, 5
Without terminal box: 0

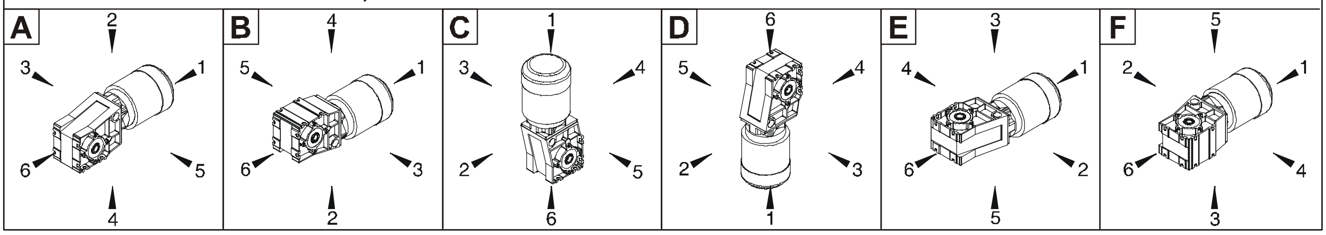


GKR

Solid shaft: 3, 5, 3+5
Hollow shaft: 0
Hollow shaft with shrink disk: 3, 5

Flange: 3, 5, 3+5
Without flange: 0

Terminal box: 2, 3, 4, 5
Without terminal box: 0



Nameplate

Geared motors

Field	Contents				Example
1	Assembly plant/country				Lenze EXTERTAL/Germany
2	Type	Drive size	Pos. of system modules	Mounting position	GFL05-2 M HCR 080-32 004 B
3		Ratio	Lubricant		i=58.667 CLP460
4		Torque M_2 in Nm	Speed n_2 in 1/min (frequency in Hz)		295 Nm 24 /min (50 Hz)
5	Order number			ID number	GT/40000027 00500038
6	Customer requirements			Year and week of manufacture	X-XXX-XX-XXXX 0203

Gearbox

Field	Contents				Example
1	Assembly plant/country				Lenze EXTERTAL/Germany
2	Type	Drive size	Pos. of system modules	Mounting position	GFL05-2 N HCK 1C 004 B
3		Ratio	Lubricant		i=27,524 CLP460
4		Torque M_2 in Nm	Reference speed 1/min		260 Nm (1400 /min)
5	Order number			ID number	GT/40000035 00500050
6	Customer requirements			Year and week of manufacture	X-XXX-XX-XXXX 0203

What is new / what has changed in the Operating Instructions?

Material number	Edition	Important	Contents
00 390 375	1.0 11/95 TD09	1st edition	1st edition for pilot series
00 393 076	1.0 03/97 TD09	1st edition Replaces 386 569	Completely revised
00 393 076	1.0 04/00 TD09	1st edition Replaces 391 707	Addendum to Ch. 4.2.1 Preparation Updated illustrations for Ch. 4.2.7 Gearbox with ventilation
00 407 986	1.0 12/00 TD09	1st edition Replaces 407 712	Completely revised Revised product key and position of system modules Supplement with gearbox 03 Ch. 4.2.11 Gearbox with reservoir in mounting position C Ch. 4.2.11.1 Spare parts list with reservoir Changes of lubricant quantities
00 425 604	1.0 08/01 TD09	1st edition Replaces 414 547	Ch. 4.2.8 Assembly of shrink disk cover: new Ch. 4.2.9 Assembly of jet-proof hollow shaft cover: new Supplement with GKR 05
00 425 604	2.0 11/01 TD09	2nd edition Replaces 1st edition	Changes of lubricant quantities
00 460 708	1.0 12/02 TD09	1st edition Replaces 425 505	Changes of nameplates Supplement with GKR 06 Change of company's name
00 460 708	2.0 02/03 TD09	2nd edition Replaces 1st edition	Addendum - Warning in Ch. 4
00 476 711	1.0 10/03 TD09/TD27	1st edition Replaces 460 114	Changes: Product key and position of system modules Addendum with Ch. 6.2.1, Roller bearing grease and Ch. 6.2.2, Lubricant table

Contents

Contents

1 Preface and general information

1.1 How to use these Operating Instructions

- These Operating Instructions are intended for safety-relevant operations on and with the gearboxes G☐☐. They contain safety instructions which must be observed.
- All personnel working at and with the gearboxes G☐☐ must have the Operating Instructions available and observe the information and notes relevant for them.
- The Operating Instructions must always be complete and perfectly readable.

1.1.1 Terminology used

Gearbox

For ”gearbox of the product group G☐☐” the term ”gearbox” will be used in the following text.

Drive system

For drive systems with gearboxes G☐☐ and other Lenze drive components the term ”drive system” will be used in the following text.

1.2 Scope of supply

- The drive systems are combined individually according to a modular design. The scope of supply can be obtained from the pertinent papers.
- After receipt of the supply, check immediately whether it corresponds with the accompanying papers. Lenze does not grant any warranty for subsequent claims. Claim for
  – visible transport damages immediately to the forwarder.
  – visible deficiencies / incompleteness immediately to the responsible Lenze subsidiary / agency.

1.3 Lenze drive systems

1.3.1 Labelling

- Lenze drive systems are uniquely designated by the content of their nameplates.
- Manufacturer:
  Lenze Drive Systems GmbH
  Postfach 10 13 52
  D-31763 Hameln

1.3.2 Application as directed

- Lenze drive systems
  – are intended for use in machinery and plant.
  – must only be used for the purposes ordered and confirmed.
  – must only be operated under the ambient conditions prescribed in these operating instructions.
  – must not be operated beyond their corresponding power limits.

Any other use shall be deemed inappropriate!

1.3.3 Legal regulations

Liability

- The information, data, and notes in the operating instructions met on the state of the art at the time of printing. Claims referring to drive systems which have already been supplied cannot be derived from the information, illustrations, and descriptions.
- We do not accept any liability for damage and operating interference caused by:
  – inappropriate use
  – unauthorised modifications to the drive system
  – improper working on and with the drive system
  – operating mistakes
  – disregarding the operating instructions

Warranty

- Conditions of warranty: see terms of sale and delivery of Lenze Drive Systems GmbH.
- Warranty claims must be made to Lenze immediately after detecting the deficiency or fault.
- The warranty is void where liability claims cannot be made.

2 Safety instructions

2.1 Personnel responsible for safety

Operator

- An operator is any natural or legal person who uses the drive system or on behalf of whom the drive system is used.
- The operator or his safety officer must ensure,
  – that all relevant regulations, instructions and legislation are observed.
  – that only qualified personnel work with and on the drive system.
  – that the personnel have the operating instructions available for all corresponding operations.
  – that non-qualified personnel are prohibited from working with and on the drive system.

Skilled personnel

Skilled personnel are persons who - because of their education, experience, instructions, and knowledge about corresponding standards and regulations, rules for the prevention of accidents, and operating conditions - are authorised by the person responsible for the safety of the plant to perform the required actions and who are able to recognise potential hazards.
(See IEC 364, definition for skilled personnel)

2.2 General safety information

- This safety information is not claimed to be complete. In case of questions and problems, please contact your Lenze representative.
- At the time of delivery the drive system meets the state of the art and ensures basically safe operation.
- The drive system is a source of danger for persons, for the drive system itself, and for other material assets of the operator, if
  – unqualified personnel works with and on the drive system.
  – the drive system is used inappropriately.
- The drive systems must be designed such that they perform their functions after proper installation and with application as directed in fault-free operation and that they do not cause hazards for persons. This also applies for their interaction with the complete plant.
- Make sure by appropriate measures that in case of failure of the drive system no material damage is caused.
- Operate the drive system only when it is in a proper state.
- Retrofittings, modifications, or redesigns of the drive system are basically prohibited. Lenze must be contacted in all cases.

2.3 Layout of the safety instructions

All safety instructions given in this documentation feature the same layout:

Icon (characterises the type of danger)

Signal word! (characterises the severity of danger)

Note (describes the danger and gives information about how to prevent dangerous situations)

Icon	Signal word		Consequences if the safety instructions are disregarded
	Signal word	Meaning	
Hazardous electrical voltage General danger	**Danger!**	**Impending danger to persons**	Death or severe injuries.
	Warning!	**Potential, very hazardous situation for persons**	Death or severe injuries.
	Caution!	**Potential, hazardous situation for persons**	Light or minor injuries
STOP	**Stop!**	**Potential damage to material**	Damage of the drive system or its environment
	Tip!	**Useful advice** If you observe it, handling of the drive system is made easier.	

3 Technical data

- The most important technical data are indicated on the nameplate (for Design and Contents see page 3).
- Further technical data are listed in the product catalogues.

3.1 Product features

3.1.1 Design

Drive systems have a modular design.

They consist of:

- Reduction gearboxes
  (helical gearboxes, low-profile gearboxes, helical-worm gearboxes and helical-bevel gearboxes)
- Variable speed drives
- Motors

3.1.2 Mode of operation

- Torque and speed conversion

<table>
<tr><th>Product family</th><th>Input stage</th><th>1st stage</th><th>2nd stage</th><th>3rd stage</th></tr>
<tr><td>Helical gearbox</td><td rowspan="3">Helical</td><td rowspan="5">Helical</td><td rowspan="2">Helical</td><td>---</td></tr>
<tr><td>Low-profile gearbox</td><td>---</td></tr>
<tr><td>Helical bevel gearbox</td><td rowspan="2">Bevel</td><td>Helical</td></tr>
<tr><td>Bevel gearbox</td><td>---</td><td>---</td></tr>
<tr><td>Helical-worm gearbox</td><td>Helical</td><td>Worm</td><td>---</td></tr>
</table>

Tab. 1 Torque and speed conversion of the gearboxes

- The reaction torque must be supported by foot mounting, flange mounting, or torque plate.

3.2 Transport weights

Gearbox size	Geared motors Motor size					
	063-□□	071-□□	080-□□	090-□□	100-□□	112-□□
G□□03	< 10	< 10				
G□□04	< 30	< 30	< 40	< 50		
G□□05	< 50	< 50	< 60	< 60	< 70	
G□□06	< 70	< 70	< 80	< 90	< 100	< 125
G□□07		< 125	< 125	< 150	< 150	< 175
G□□09		< 200	< 200	< 225	< 225	< 250
G□□11			< 350	< 375	< 375	< 400
G□□14				< 625	< 650	< 650

Gearbox size	Geared motors Motor size					Gearboxes
	132-□□	160-□□	180-□□	200-□□	225-□□	
G□□04						< 30
G□□05						< 50
G□□06						< 70
G□□07	< 200	< 250				< 150
G□□09	< 275	< 325	< 475	< 550		< 250
G□□11	< 425	< 450	< 600	< 700	< 850	< 400
G□□14	< 700	< 750	< 850	< 950	< 1100	< 625

Tab. 2 Weights in kg

3.3 Application conditions

3.3.1 Temperatures

The permissible temperature range is determined by the following:

- The specification of the lubricant considering the oil temperature expected during operation (see chapter 6.1 and nameplate page 3).
- The thermal class of the motor considering the motor temperature expected during operation (see nameplate and/or Operating Instructions of the motor).

The operating temperature is determined by the power loss, the ambient temperature and the cooling system!

3.3.2 Ambient conditions

- Gearboxes are protected against dust and spray water.
- Motors according to their enclosure (see nameplate and/or Operating Instructions of the motor).
- Ambient media - especially chemically aggressive - can destroy shaft seals and coatings (plastic). Abrasive media endanger shaft seals.

4 Installation

Danger!

Only transport the drive with transport equipment or hoists which are suitable for this load (see transport weights, chapter 3.2). Ensure a safe fixing. Avoid shocks!

The motors attached to the gearbox are partially equipped with eyebolts. These are **exclusively** determined for motor/gearbox mounting and dismounting and must **not** be used for the complete geared motor!

4.1 Storage

If you do not install the gearbox immediately, ensure appropriate conditions of storage.

- Up to one year:
  Without special measures in dry, dust-free rooms which are protected against sunlight.
  – Store the gearbox with ventilation such that the breather screw is located at the top.
  – Shafts and bright surfaces are delivered with protection against corrosion.
  – With motors with condensation bore (option), the plug screw must be removed (see chapter 4.2.1).
- More than a year:
  Contact the factory.

4.2 Installation

4.2.1 Preparation

- Thoroughly remove anticorrosion agents from output shafts and flange faces.
- As standard, geared motors providing the option “Condensation bore” are delivered with plug screws. Please ensure, that the plug screws are removed when the geared motor is installed or stored. Depending on the mounting position, the plug screws are at the bottom of the motor.

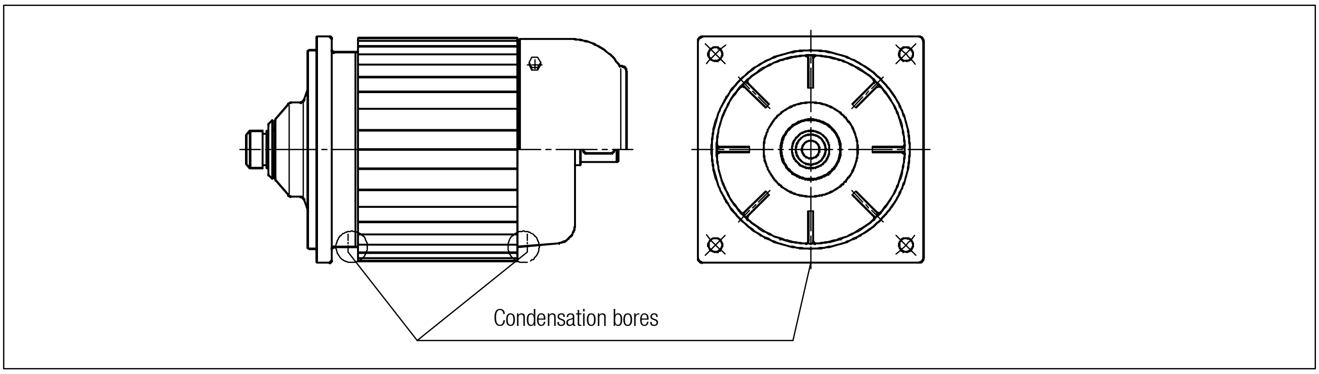


Fig. 1 Motor with condensation bores

4.2.2 General information about the assembly of drive systems

- Take safety measures prior to any operation:
  - Disconnect the machine from the mains, ensure standstill of the machine and avoid any machine movement.
  - Check the proper state of the drive system. Never install and set up damaged drive systems.
  - Check the combination of drive function and machine functions. Check the direction of rotation.
- The mounting surfaces must be even, without torsion, and free from vibration.
- Align the drive system on the mounting surfaces exactly with the machine shaft to be driven.
  - Ensure that the assembly is without torsion, to avoid additional load.
  - Compensate for minor misalignments by using suitable flexible couplings.
- Support the reaction torque appropriately.
- Fixings of accessories and attachments must be secured against loosening. We recommend that screw connections are glued.

STOP

Stop!
The lubricant quantity of the gearbox is depends on the mounting position. Therefore it is absolutely necessary to mount the gearbox as indicated on the nameplate to avoid any damage (for mounting positions see chapter 6.2 and nameplate, page 3).
For speeds < 200 min^{-1}, please contact Lenze!

4.2.3 Assembly of transmission elements on solid shafts

- Draw the transmission elements onto the output shaft only by using the centering thread.

STOP

Stop!
Shocks and blows to the shafts damage the roller bearings.

4.2.4 Attachment of motors to gearboxes with bearing housing (input design N)

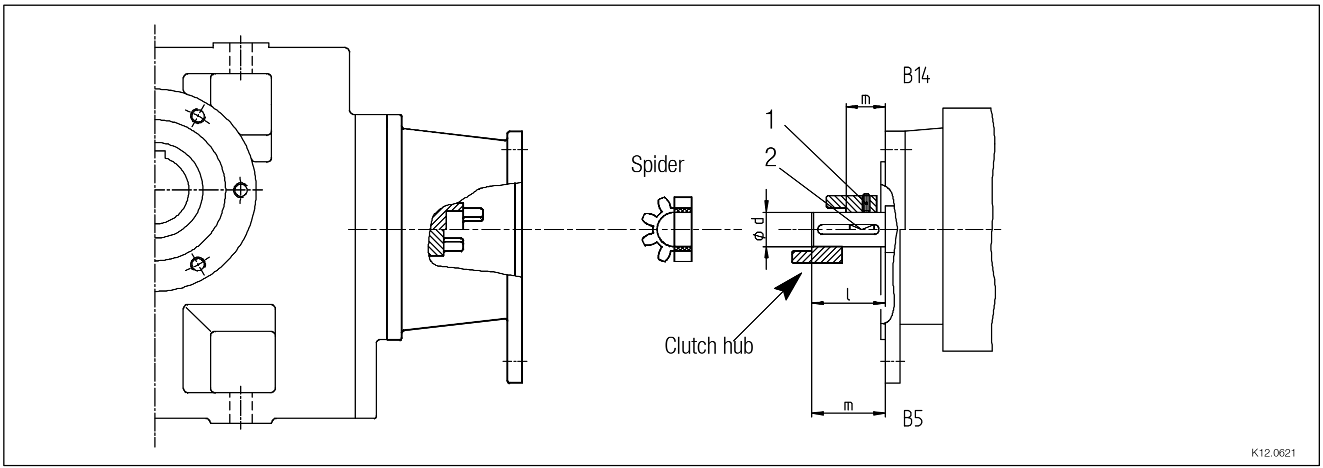


Fig. 2 Input side design N

Drive size	Motor shaft d [mm]	Motor shaft max. l [mm]	Assembly dimension m [mm]	Standard hub Fixing screw Thread [mm]	Clamping hub Thread [mm]	Clamping hub Torque [Nm]	Key 1) DIN 6885/1 [mm]	Clamping ring hub Thread [mm]	Clamping ring hub Torque [Nm]
1A	11	23	23	M4	M3	1.34	*	M3	1.34
1B	14	30	30	M4	M3	1.34		M3	1.34
2B	11	23	23						
1C	19	40	25	M5	M6	10.5	B 6 x 6 x 16	M4	2.9
2C	14	40	25				B 5 x 5 x 16		
3C	14	40	25						
4C	14	40	25						
1D	24	50	50	M5	M4	2.9	*	-	-
2D	19	40-50	50		M6	10.5		M4	2.9
1E	28	30-60	30	M5	M6	10.5	B 8 x 7 x 18	M5	6
2E	24	30-60	30						
3E	19	30-60	30				B 6 x 6 x 18		
1F	28	30-60	30	M5	M6	10.5	B 8 x 7 x 18	M5	6
2F	24	30-60	30						
1G	38	80	80	M6	M8	25	*	M5	6
2G	28	60	60						
3G	38	80	80						
1H	42	110	110	M8	M10	69	*	-	-
2H	48	110	110						
3H	38	80	80					M8	35
1K	55	110	110	M8	M10	69	*	-	-
2K	60	140	140						

Tab. 3 Attachment of motors to gearboxes with bearing housing
* Use original key for the motor
1) Key for standard and clamping hub

4.2.4.1 Assembly of standard / clamping hub

1. Assemble motor key (2).
   – Assemble enclosed key for drive sizes ☐C, ☐E, ☐F.
2. Push coupling hub on the motor shaft, observe dimension m.
3. Secure coupling hub against axial movement using the fixing screw or clamping screw (1).
4. Lay spider into the coupling claw at the gearbox side.
5. Align claws of the motor-side coupling hub with its counterpart.
6. Slowly push motor, and bolt on to the gearbox flange.

4.2.4.2 Assembly of clamping ring hub

1 Clamping ring hub
2 Clamping ring
3 Clamping screws (DIN912)

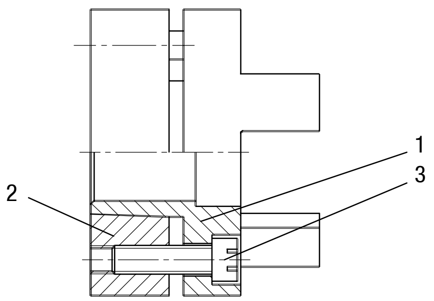


Fig. 3 Coupling

1. Grease the contact surfaces of the motor shaft using a thin-bodied oil, e. g. “Castrol 4 in 1” or “Klüber Quitsch Ex”!

> STOP
>
> **Stop!**
>
> Do <u>not</u> use oil or grease with molybdenum-disulphid or high-pressure sets as well as grease paste!

2. Push the clutch hub over the motor shaft, mounting dimension “m” (see Fig. 2 and Tab. 3) must be observed.
3. Align the hub and tighten the clamping screws until they have contact.
4. Tighten the screws evenly and crosswise using a torque wrench until the indicated tightening torque is reached at all screws.
5. Lay spider into the coupling claw at the gearbox side.
6. Align claws of the motor-side coupling hub with its counterpart.
7. Slowly push motor, and bolt on to the gearbox flange.

4.2.4.3 Disassembly of clamping ring hub

1. Loosen the clamping screws evenly one after the other.
2. Remove the screws next to the forcing threads and screw them into the other threads until they have contact.
3. Tighten the screws in the forcing threads crosswise and step-by-step so that the clamping ring is pushed from the tapering clamping ring hub.
4. Clean and grease all contact surfaces including threads and head of the clamping screws before reassembly.

4.2.4.4 General

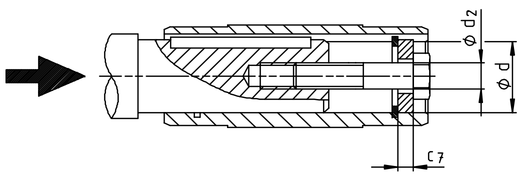
Tip!

Standard, clamping and clamping ring hubs are maintenance-free.We recommend to check the star-shaped spider and system components when inspecting the drive.

4.2.5 Attachment of gearboxes with hollow shaft and keyway

Assembly

1. Draw the gearbox with hollow shaft onto the machine shaft to be driven:
   – Use a suitable assembly paste.
   – Take up forces only via the hollow shaft, and not via gearbox housing.
2. Secure the gearbox axially:
   – The hollow shaft has snap ring grooves for axial securing (Fig. 4). Parts which are used to fix the shaft are not included in the scope of supply.



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Auxiliary tool (recommended dimensions)		
Ø d^{H7}	Ø d_2	c_7
18 20	M6	4
25 30	M10	5 6
35	M12	6
40 45	M16	8
50 55	M16 M20	10
60 70	M20	12
80	M20	14
100	M24	16

Tab. 4 Dimensions in [mm]

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Dismantling

1. Remove axial securing of the gearbox from the hollow shaft.
2. Remove the gearbox from the motor shaft using an appropriate auxiliary tool (Fig. 4).

Tip!
Use suitable extractors for bevel gearbox GKR!

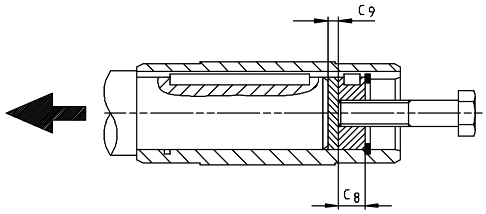


Fig. 4 Dismantling of gearboxes with hollow shaft using the auxiliary tool, not for GKR

Auxiliary tool (recommended dimensions)		
Ø d^{H7}	c_8	c_9
18 20	6	3
25 30	10	3
35	12	3
40 45	16	4
50 55	16 20	5
60 70	20	5
80	20	6
100	24	8

Tab. 5 Dimensions in [mm]

4.2.6 Attachment of gearboxes with hollow shaft with shrink disk

A Outer ring

B Inner ring

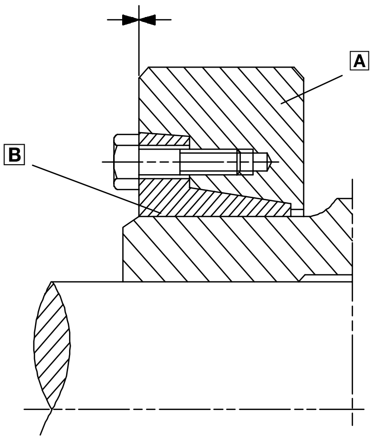


Fig. 5 Hollow shaft with shrink disk

Stop!

Do not dismantle new shrink disk.

Never tighten retaining screws before the machine shaft is pushed in.Protect the shrink disk against contact while being in operation by appropriate measure (e.g. cover).

Assembly:

1. Degrease hollow shaft bore and machine shaft.
2. Push drive onto machine shaft.
3. Tighten retaining screws using torque wrench consecutively in several turns. Observe tightening torques:

Hollow shaft bore [mm]	20	25	30	35	40	50	60	65	80	100
Torque [Nm]	12	30	30	30	30	30	59	70	59	100

4. Check correct fixing:
   – The shrink disk is mounted correctly and fixed when the faces of the outer ring and the inner ring are aligned. Minimum misalignments are permissible.

Dismantling:

1. Loosen the clamping screws evenly one after the other.
2. Extract outer ring, if necessary.
3. Remove the drive from the motor shaft.
4. Dismantle the shrink disk only for cleaning purposes. Afterwards, lubrication is required.

4.2.7 Gearboxes with breathers

When using gearbox sizes 03, 04 and GKR05, 06 it is not necessary to provide special ventilation.

Breather elements are included for gearbox sizes 09 to 14. When using these gearbox sizes in mounting position C, we recommend the use of an oil compensator.

Gearboxes supplied with a breather element bear a corresponding label on the gearbox.

STOP

Stop!

Ensure breathing before the first set-up!

- Position the gearbox according to the indications on the nameplate (see nameplate and chapter 6.2).
- With gearbox types GST□□-3, GFL□□-3, GSS□□-3 and GKS□□-4, the pre-stage is separately ventilated.
- Remove the sealing plug as indicated for the corresponding mounting position (see the following drawing) and replace it by the supplied breather element.

For breather elements with brackets:

- Mount the bracket onto the gearbox as indicated.
  Caution! The inner thread must be at top after the mounting (according to the mounting position). Screw the breather element into the bracket.

4.2.8 Mounting Instructions for shrink disk cover

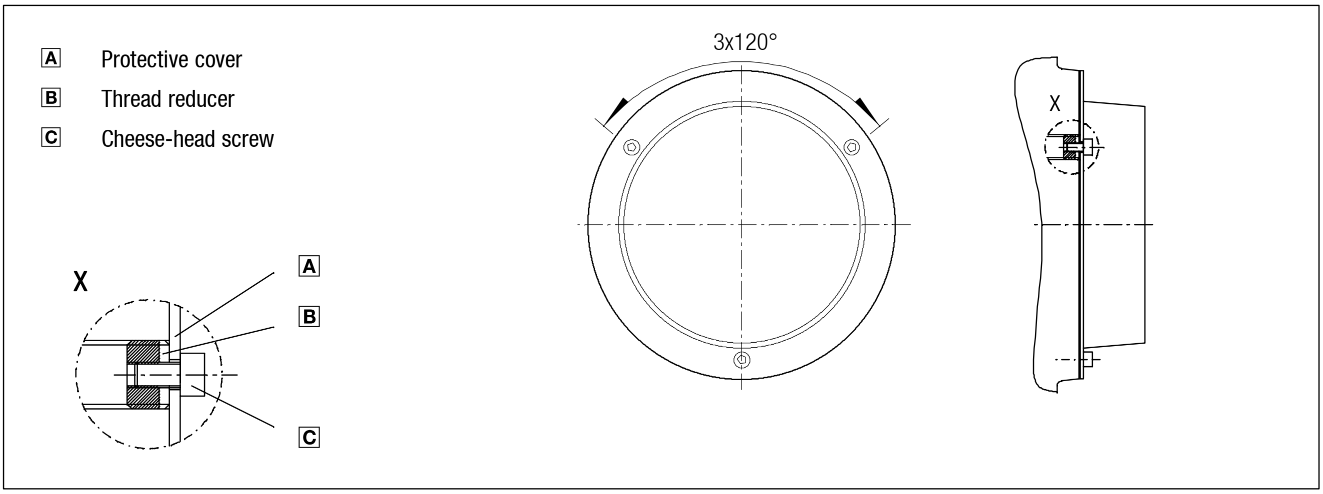


Fig. 6

1. Screw the three thread reducers into the flange, ensure **even** surfaces and stagger the reducers by 120°.
2. Mount the protective cover to the flange using the three cheese-head screws.

4.2.9 Mounting Instructions for jet-proof hollow shaft cover

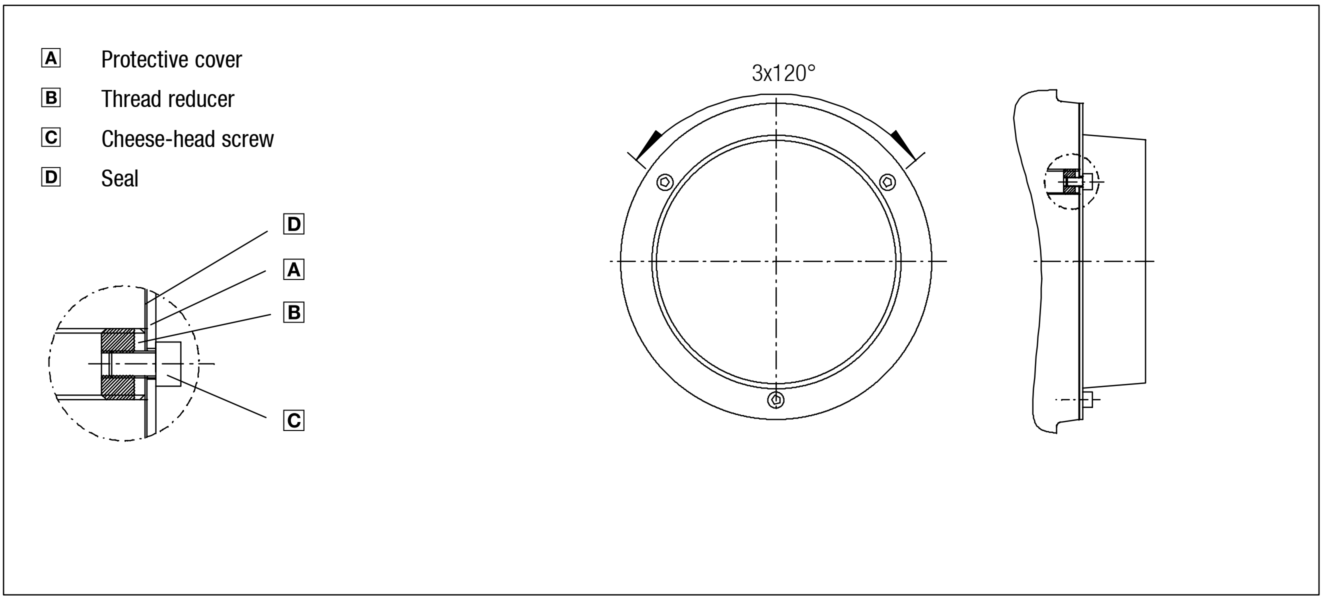
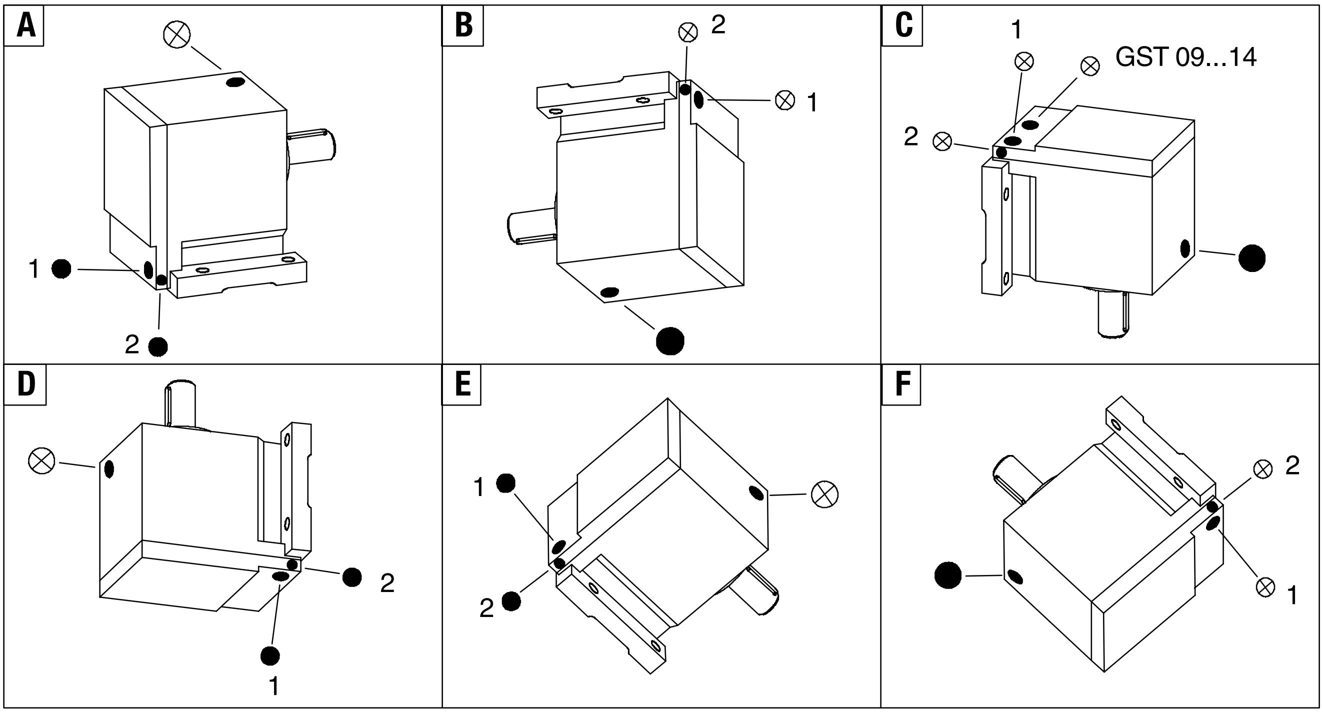


Fig. 7

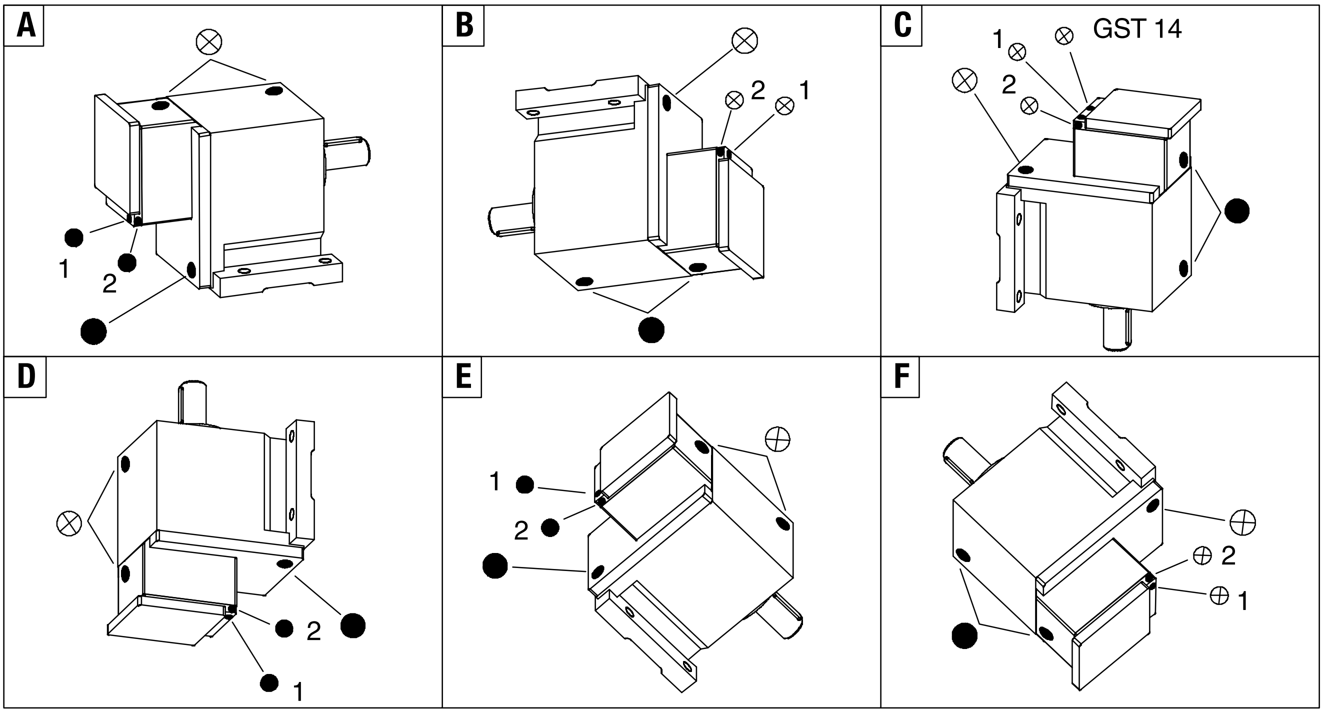
1. Screw the three thread reducers into the flange, ensure **even** surfaces and stagger the reducers by 120°.
2. Insert seal between flange and protection cover.
3. Mount the protection cover to the flange using the three cheese head screws.

4.2.10 Position of the breathing, oil-filler and oil-drain plug

4.2.10.1 Helical gearbox GST 05...09-1 and GST 05...14-2



4.2.10.2 Helical gearbox GST 05...14-3



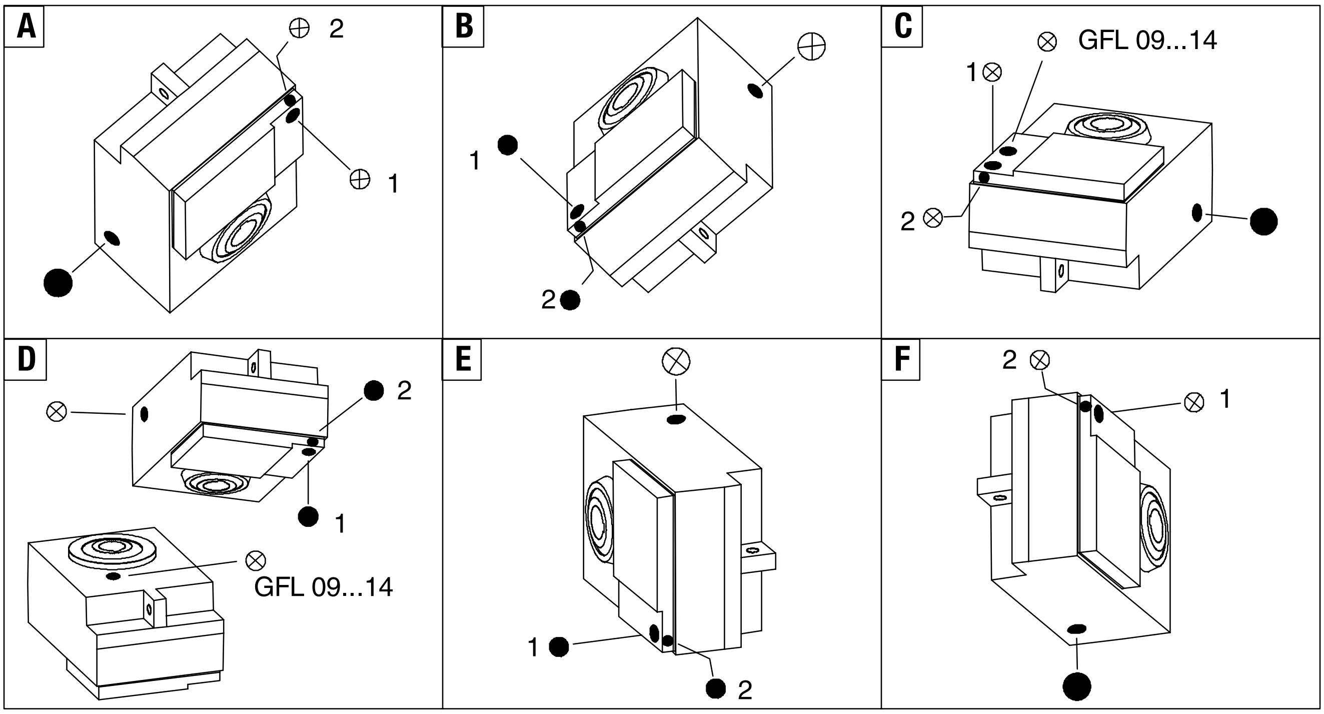
A...F	Mounting position	**Pos. 1 or 2**	acc. to version
⊗	Breathing / oil-filler plug	●	Oil-drain plug

Installation

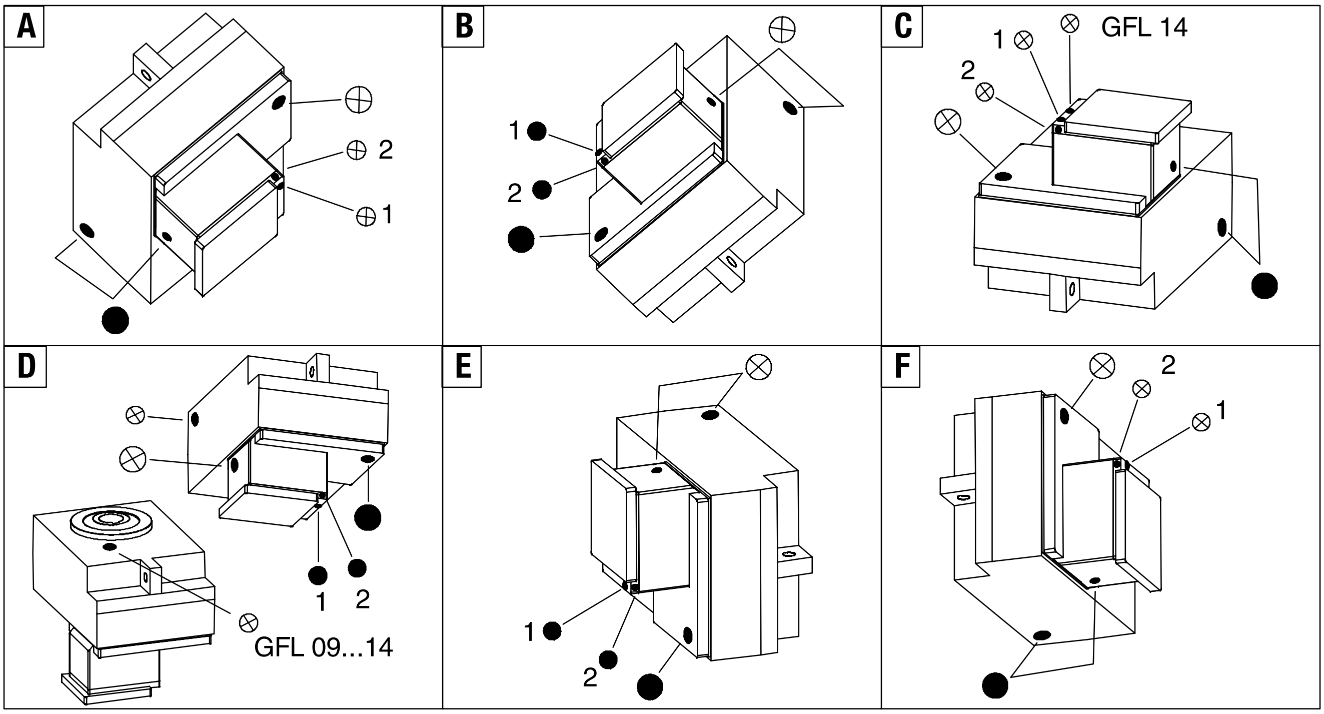
Installation

Position of the breathing, oil-filler and oil-drain plug

4.2.10.3 Low-profile gearbox GFL 05...14-2

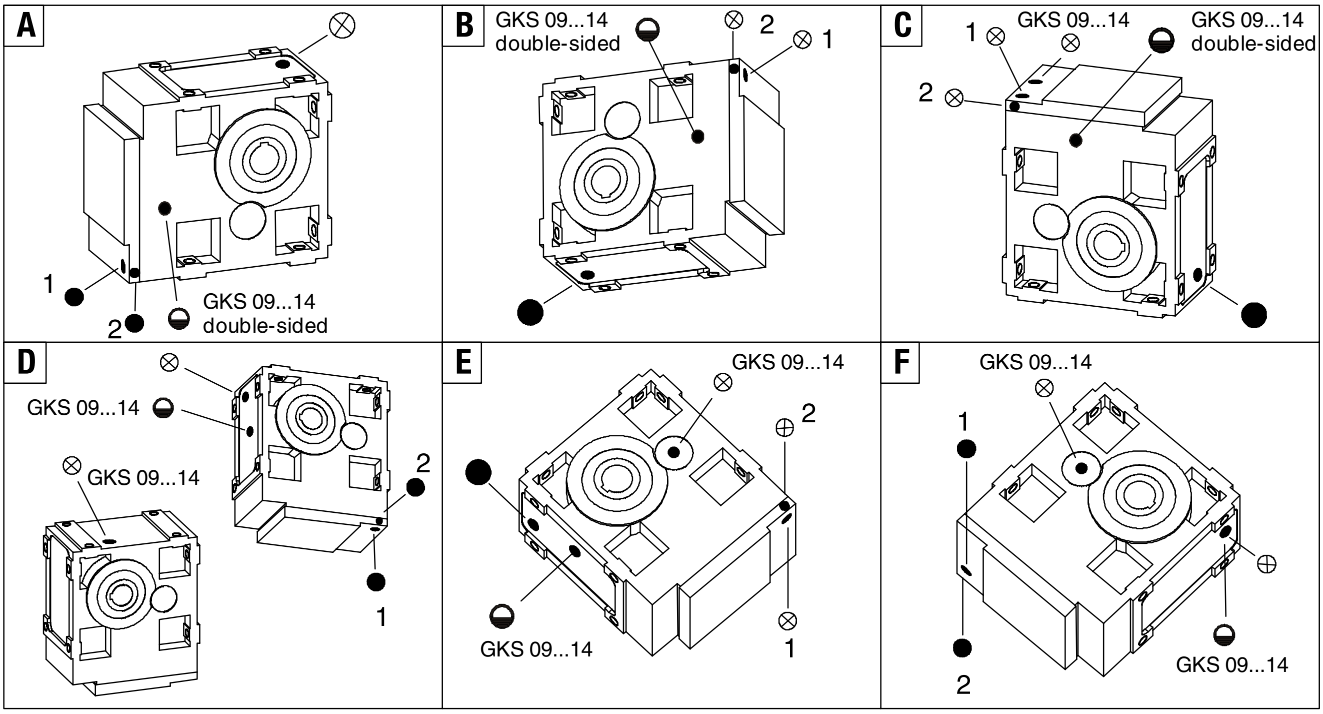


4.2.10.4 Low-profile gearbox GFL 05...14-3

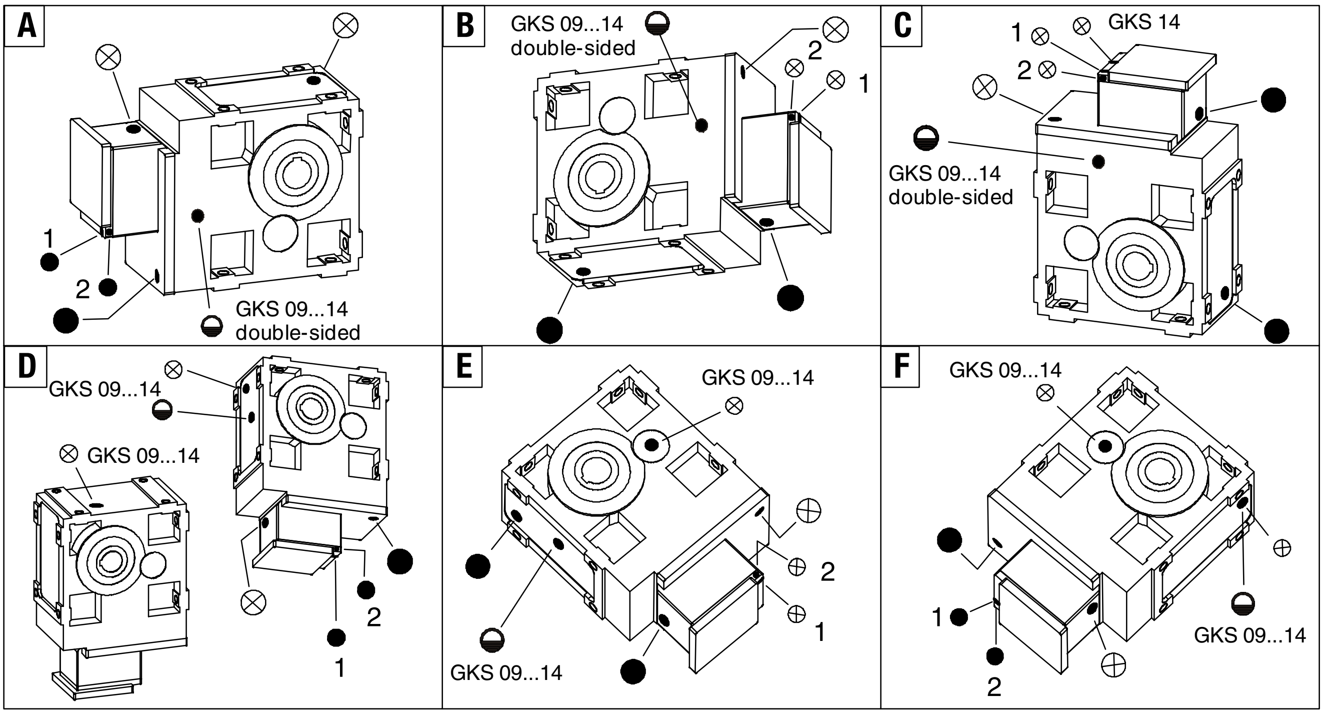


A...F	Mounting position	**Pos. 1 or 2**	acc. to version
⊗	Breathing / oil-filler plug	●	Oil-drain plug

4.2.10.5 Helical bevel gearbox GKS 05...14-3



4.2.10.6 Helical bevel gearbox GKS 05...14-4



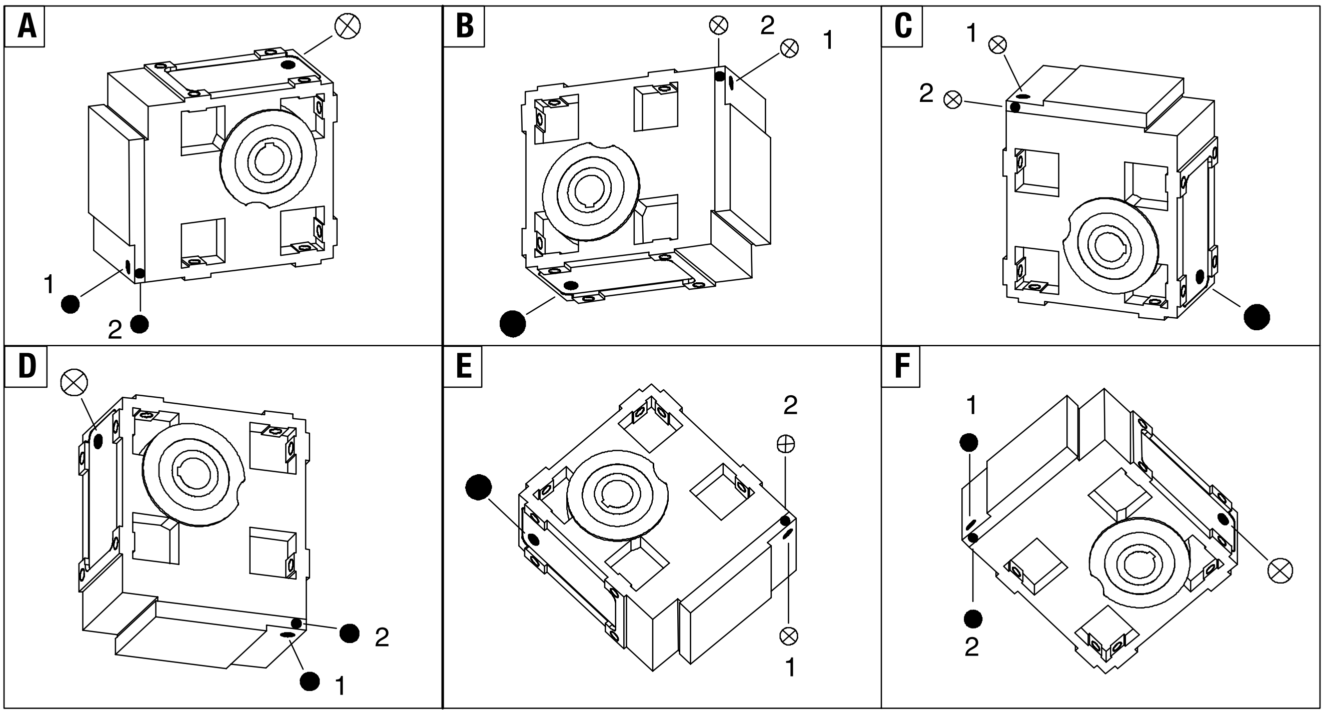
A...F	Mounting position	⊗	Breathing / oil-filler plug	◒	Oil-control plug
		●	Oil-drain plug	**Pos. 1 or 2**	acc. to version

Installation

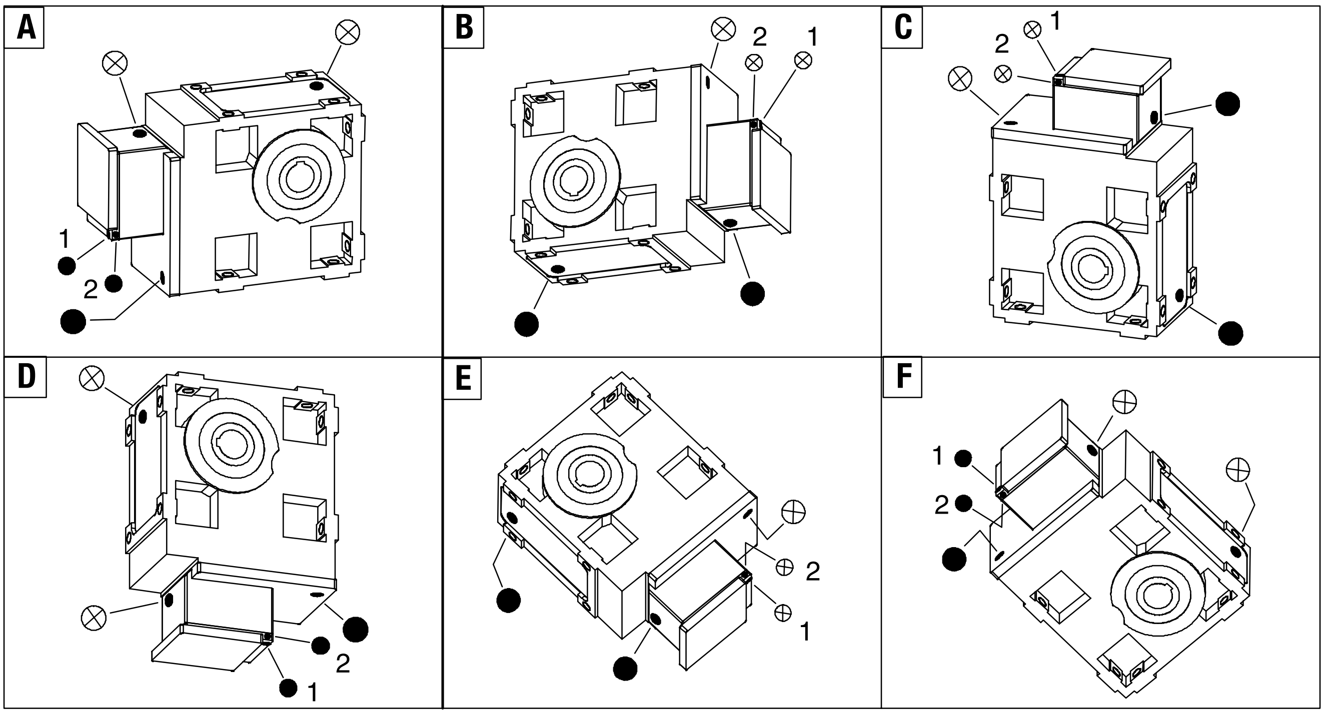
Installation

Position of the breathing, oil-filler and oil-drain plug

4.2.10.7 Helical-worm gearbox GSS 05...07-2

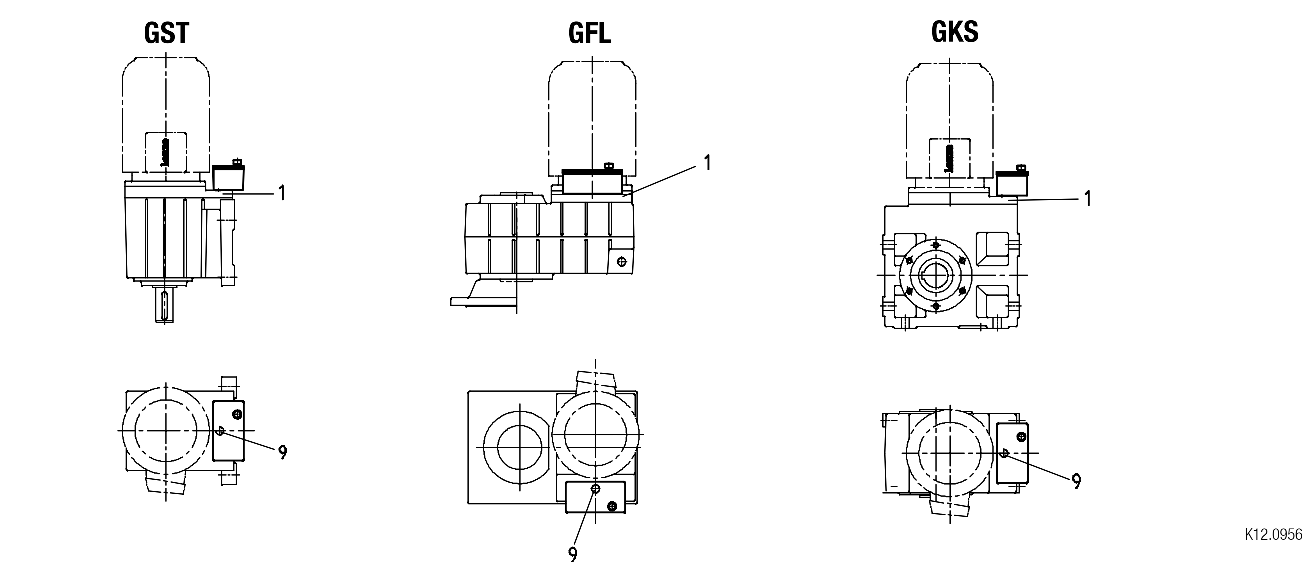


4.2.10.8 Helical-worm gearbox GSS 05...07-3



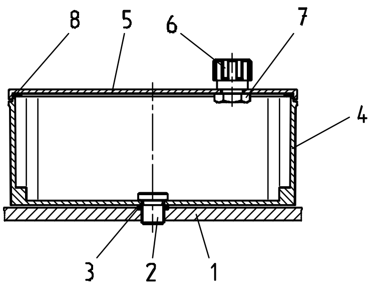
A...F	Mounting position	**Pos. 1 or 2**	acc. to version
⊗	Breathing / oil-filler plug	●	Oil-drain plug

4.2.11 Gearbox with compensation container for mounting position C



4.2.11.1 Spare-parts list compensation container

1 Intermediate cover
2 Fixing screw
3 Seal rings
4 Housing
5 Lid
6 Breathing filter
7 Hexagon screw
8 Seal



Assembly

1. Mount gearbox in position (motor on top).
2. Remove the plug from the intermediate cover (9).
3. Mount housing (4) to the intermediate cover (instead of the plug).

Caution! Insert two seals between counterbores of intermediate cover and housing.

4. Use hexagon screw to bolt the breathing filter to the lid.
5. Mount lid and seal to the cover.

Stop!

For transport the compensation container must be removed and the plug must seal the gearbox (pos. 9).

4.3 Electrical connection

Danger!
Electrical connections must only be made by skilled personnel!

4.3.1 Motor connection

To correctly connect the motor options, e.g. Lenze spring-loaded brakes, please observe:

- the notes in the terminal box of the motor
- the notes in the operating instructions of the motor
- the technical data on the motor nameplate.

4.3.2 Motor options

To correctly connect the motor options, e. g. Lenze spring-loaded brakes, please observe:

- the notes in the corresponding terminal box
- the notes in the corresponding operating instructions
- the technical data on the corresponding motor nameplate.

5 Commissioning and operation

Stop!
The drive may only be commissioned by skilled personnel!

5.1 Before you start

Check:

- Is the mechanical fixing o.k.?
- Are the electrical connections o.k.?
- Are all rotating parts and surfaces, which may become hot, protected against contact?
- For gearboxes with ventilation (chapter 4.2.7):
  Ensure ventilation!

5.2 During operation

Stop!
The helical-worm gearboxes reach their full performance only after a short running-in period of 24...48 hours at rated torque!

- During operation, check the drive periodically and take special care of:
  – unusual noises or temperatures
  – leakages
  – loose fixing elements
  – the condition of the electrical cables.
- If any interference should occur, proceed according to the troubleshooting list in chapter 7. If the interference cannot be eliminated, please contact the Lenze Service.

6 Maintenance

Lenze gearboxes and geared motors are filled with a drive and design-specific lubricant filling upon delivery. This original filling corresponds to the lubricant listed in the column of the respective gearbox type from Lenze. The important factors for the lubricant filling at ordering are the mounting position and the design.

Tip!

Gearboxes of size 03 and 04 are lubricated for life. Because of the minimum thermal load it is not necessary to replace the lubricant.

When changing the lubricant, Lenze recommends also changing the grease packing of the bearings and replacing the radial shaft seal rings!

6.1 Maintenance intervals

- The mechanical power transmission system is free of maintenance.
- Gearboxes as of size 05 require frequent lubricant replacement.
  - The type of lubricant is indicated on the nameplate. Replace the lubricant only by the same type.
  - Maintenance intervals: see Tab. 6 "List of lubricants".
- Shaft seal rings:
  - The service life depends on the ambient conditions.
  - Replace seals in case of leakage to avoid consequential damage.

Stop!

For drive systems: Also observe the maintenance intervals for the other drive components!

Lubricants				Changing interval
Type	Specification	Ambient temperature	Note	
CLP 460	Oil on mineral basis with additives	0 °C ... + 40 °C		16.000 operating hours - no longer than three years (oil temperature 70...80 °C)
CLP PG 220 CLP PG 460 CLP PG 680	Oil on synthetic basis (polyglycol)	-20 °C ... + 40 °C	Do not mix with mineral oils!	25.000 operating hours - no longer than three years (oil temperature 70...100 °C)
CLP HC 220 (H1)	Food-compatible oil on synthetic basis	-20 °C ... + 40 °C	Approval to USDA-H1	16.000 operating hours - no longer than three years (oil temperature 70...80 °C)
CLP E 320	Biodegradable diester oil on synthetic basis	-20 °C ... + 40 °C	Water pollution class 0	16.000 operating hours - no longer than three years (oil temperature 70...80 °C)
CLP HC 320	Oil on synthetic basis (synthetic carbon hybrid)	-25 °C ... + 50 °C	Can be mixed with residuary mineral oil	25.000 operating hours - no longer than three years (oil temperature 70...80 °C)
CLP HC 46	Oil on synthetic basis (synthetic carbon hybrid)	-40 °C ... 0 °C	Good cold-flow behaviour	25,000 operating hours - no longer than three years

Tab. 6 Overview of lubricant change

6.2 Maintenance operations

6.2.1 Roller bearing grease

The roller bearings of motors and gearboxes from Lenze are filled with the grease listed below:

	Ambient temperature	Manufacturer	Type
Gearbox roller bearing GST, GFL, GKS, GKR	-30...+50°C -30...+80°C -40...+60°C	Fuchs Klüber Klüber	Renolit H 443 Petamo 133N Microlube GHY 72
Gearbox roller bearing GSS	-30...+80°C -15...+60°C	Klüber Klüber	Petamo 133N Klüberplex BE 11-462
Motor roller bearing	-30...+70°C -40...+80°C	Lubcon Klüber	Thermoplex 2TML Asonic GHY 72
Specialty grease for gearbox roller bearing			
Observe low-temperature oils, critical starting performance at low temperatures!	-40...+80°C	Klüber	Asonic GHY 72
Biodegradable oil (lubricant for forestry, agriculture and water economy)	-40...+50°C	Fuchs	Plantogel 0120S

The following lubricant quantities are required:

- For fast-running bearings (motor and drive end of gearbox): fill approx. one-third of the hollow space between rolling bodies with grease.
- For slow-running bearings (in gearbox and driven side of gearbox): fill approx. two-thirds of the hollow space between roller bearings with grease.

Tip!

Please note that the recommendation of a lubricant/grease or its listing in a Lenze lubricant table does not mean that Lenze is liable for these lubricants or damages resulting from incompatibilities of materials used.

6.2.2 Table of lubricants

The lubricants listed in the lubricant table are permissible for Lenze drives. Specialty lubricants must be used, for example, for long-term storage or special operating conditions. These corresponding lubricants are available at a surcharge.

For the lubricant selection the following legend for the lubricant table must be observed:

CLP => Mineral oil

CLP PG => Polyglycol oil

CLP PAO => Synthetic hydrocarbons or poly-alpha-olefin oil

CLP E => Diester oil (water pollution class WPC 1)

1) => No test results are currently available about the performance of listed lubricants for worm gearbox lubrication. If these oils are used, the permissible torque must be reduced to 80% of the catalogue values.

2) => Polyglycol oils cannot be mixed with other oil types

3) => With ambient temperatures of more than 40 °C please contact us with regard to exact application conditions!

4) => Observe critical starting performance at low temperatures! At temperatures below -25 °C, special measures are required for motor bearing and NBR shaft seal rings!
=> Observe low-temperature oil, critical starting performance!

// => Food-compatible oil (approval to USDA-H1)

≈ => Biodegradable oil (lubricant for forestry, agriculture and water economy)

* => Low-temperature oils, observe critical starting performance at low temperatures!

(grey field) => Factory-filled lubricants from Lenze

	Ambient temperature [°C] -50 0 +50		DIN 51517-3: CLP ISO 12925-1: CKC/CKD			Gearbox types GST, GFL, GKS, GKR	Gearbox types GSS
Shell	0	+40		CLP	VG 460	Omala 460	
	-25	+50 3)		CLP HC	VG 320	Omala HD 320	
	-10	+50 3)	//	CLP HC	VG 460	Cassida Fluid GL 460	
	-20	+40	//	CLP HC	VG 220	Cassida Fluid GL 220	
	-20	+40		CLP PG	VG 220		Tivela S 220 2)
	-20	+40		CLP PG	VG 460		Tivela S 460 2)
	-40	0 4)	*	CLP HC	VG 46	Cassida HF 46	
	-20	+40	//	CLP PG	VG 320		Cassida Fluid WG 320 1) 2)
	-20	+50 3)	≈	CLP E	VG 320	Omala EPB 320	Omala EPB 320 1)

	Ambient temperature [°C] min. (-50 … 0 … +50)	max.		DIN 51517-3: CLP / ISO 12925-1: CKC/CKD		Gearbox types GST, GFL, GKS, GKR	GSS
KLÜBER LUBRICATION	0	+40		CLP	VG 460	Klüberoil GEM1 460	
	-25	+50 3)	*	CLP HC	VG 320	Klübersynth EG 4-320	
	-20	+40		CLP PG	VG 680		Klübersynth GH 6-680 2)
	-20	+40		CLP PG	VG 220		Klübersynth GH 6-220 2)
	-30	0 4)	*	CLP PG	VG 32		Klübersynth GH 6-32 1) 2)
	-40	0 4)	* //	CLP HC	VG 46	Klüber Summit HySyn FG-46	
	-20	+40	//	CLP HC	VG 220	Klüberoil 4 UH1-220N	
	-20	+40	//	CLP PG	VG 320		Klübersynth UH 1-320 1) 2)
	-20	+50 3)	≈	CLP E	VG 320	Klübersynth GEM 2-320	Klübersynth GEM 2-320 1)
FUCHS	-25	+50 3)		CLP HC	VG 320	Renolin Unisyn CLP 320	
	-20	+40	≈	CLP E	VG 320	Plantogear 320 S	Plantogear 320 S 1)
	-20	+40		CLP PG	VG 460		Renolin PG 460 1) 2)
	0	+40		CLP	VG 460	Renolin CLP 460	
ARAL	-10	+50 3)	//	CLP HC	VG 460	Eural Gear 460	
	-25	+40	//	CLP HC	VG 220	Eural Gear 220	
	-20	+40		CLP PG	VG 460		Degol GS 460 1) 2)
	0	+40		CLP	VG 460	Degol BG 460	
	-25	+50 3)		CLP HC	VG 320	Degol PAS 320	
Eni	0	+40		CLP	VG 460	Blasia 460	
	-25	+50 3)		CLP HC	VG 320	Blasia SX 320	
bp	0	+40		CLP	VG 460	Energol GR-XP 460	
	-20	+50 3)		CLP HC	VG 320	Enersyn HTX 320	
Castrol	0	+40		CLP	VG 460	Alpha MW 460	
	0	+40		CLP	VG 460	Alpha SP 460	
	-20	+40		CLP PG	VG 460	Alpha PG 460 2)	Alpha PG 460 1) 2)
DEA	0	+40		CLP	VG 460	Falcon CLP 460	
	-20	+40		CLP PG	VG 460		Polydea PGLP 460 1) 2)
	-20	+50 3)	≈	CLP E	VG 320	Ergon ELP 320	Ergon ELP 320 1) 2)
Esso	0	+40		CLP	VG 460	Spartan EP 460	
	-20	+40		CLP PG	VG 460		Glycolube 460 1) 2)
	-25	+50 3)		CLP HC	VG 320	Spartan Synthetic EP 320	
Mobil	0	+40		CLP	VG 460	Mobilgear 634	
	-20	+40		CLP PG	VG 460		Mobil Glygoyle HE 460 1) 2)
	-20	+50 3)		CLP HC	VG 320	Mobilgear SHC XMP 320	
LUBCON	0	+40		CLP	VG 460	Turmogearoil 460 OM	
	-25	+50 3)		CLP HC	VG 320	Turmofluid GV 320	
	-20	+40		CLP PG	VG 460		Turmopoloil 460 EP 1)
	-20	+40		CLP PG	VG 220		Turmopoloil 220 EP 1)
	-40	0 4)	*	CLP HC	VG 46	Turmofluid GV 46	
	-20	+40	//	CLP HC	VG 220	Turmosynthoil GV 220	
	-20	+40	//	CLP PG	VG 460		Turmosynthoil PG 460 1) 2)
	-20	+50 3)	≈	CLP E	VG 320	Turmofluid Biolube CLP 320	Turmofluid Biolube CLP 320 1)
Optimol	0	+40		CLP	VG 460	Optigear BM 460	
	-25	+50 3)		CLP HC	VG 320	Optigear Synthetic A 320	
Tribol	0	+40		CLP	VG 460	Tribol 1100/460	
	-20	+40		CLP PG	VG 460		Tribol 800/460 1) 2)
	-25	+40		CLP HC	VG 320	Tribol 1510/320	
	-20	+40	//	CLP	VG 220	Food Proof 1810/220	
	-20	+50 3)	//	CLP PG	VG 460		Food Proof 1800/460 1) 2)

6.2.3 Replacing the lubricant

- Gearbox should be warm.
- Secure drive system and machine from inadvertent movement and mains connection.

Stop!

With gearbox types GST□□-3, GFL□□-3, GSS□□-3 and GKS□□-4, the pre-stage is separately lubricated–all gearbox components have to be completely drained!

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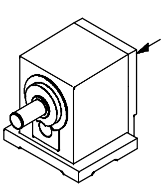
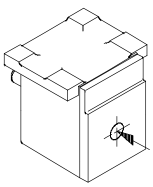
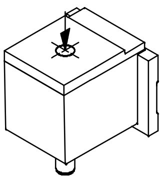
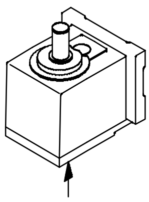
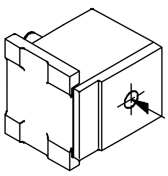
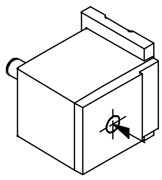
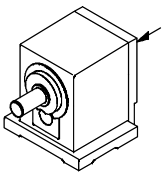
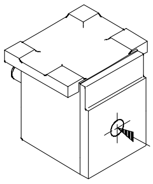
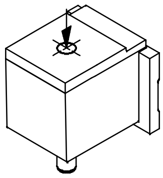
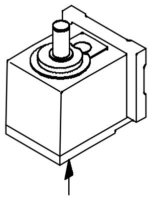
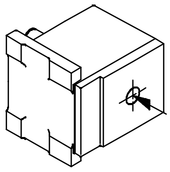
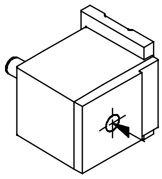
1. Drain lubricant by removing the oil drain plug (see figure on pages 20-23).
2. Insert oil drain plug with new seal.
3. Remove oil filler plug/breather element.
4. Fill up lubricant through filler.
5. Insert oil filler plug/breather element with new seal.
6. Dispose of waste oil according to the applicable regulations.

Lubricant quantity of helical gearboxes GST□□-1□ VA□/VB□						
Mounting position	A 	B 	C 	D 	E 	F 
GST04	0.1	0.37	0.2	0.3 0.35 W ≥ 1C	0.25	0.25
GST05	0.2	0.6	0.35	0.5 0.6 M ≥ 90 A ≥ 80 N ≥ 1D	0.35	0.35
GST06	0.4	1.2	0.65	0.85 1 W ≥ 1E	0.7	0.7
GST07	0.7 1.3 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.3 2.7 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.3	1.5 2.2 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.5	1.5
GST09	1.2 2.5 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	4.1 4.8 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.8	2.7 3.7 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.5	2.5

Tab. 7 Lubricant quantity in litre

Note:
➔ = Drive
A, M, N, W = Input design
90.....132 = Motor frame size
1A.....2K = Drive size

Lubricant quantity of helical gearboxes GST□□-1□ VC□						
Mounting position	A 	B 	C 	D 	E 	F 
GST04	0.1	0.3	0.15	0.3 0.35 W ≥ 1C	0.2	0.2
GST05	0.2	0.5	0.2	0.45 0.55 M ≥ 90 A ≥ 80 N ≥ 1D	0.3	0.3
GST06	0.4	1	0.45	0.85 1 W ≥ 1E	0.6	0.6
GST07	0.8 1.5 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.6 2.0 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	0.85	1.6 2.3 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.3	1.3
GST09	1.6 2.7 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	3.0 3.5 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.7	2.7 3.7 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.3	2.3
Lubricant quantity of helical gearboxes GST□□-2□; -3□ VA□/VB□						
GST03	0.2	0.2	0.2	0.2	0.2	0.2
GST04	0.35	0.5	0.45	0.55	0.3	0.3
GST05	0.7	0.85	0.75	1	0.55	0.55
GST06	1.25	1.5	1.35	1.8	1	1
GST07	2.2 2.6 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.5 2.9 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.6	3 3.7 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.7 2.1 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.7 2.1 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G
GST09	4.2 4.8 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	5.3 5.9 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	5.4	6.1 7.3 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	3.1 3.7 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	3.1 3.7 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G
GST11	8.5	9.5	10	11.5	7	7
GST14	15	18	18	20	11	11

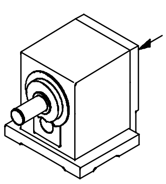
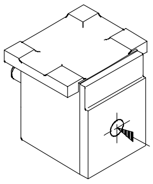
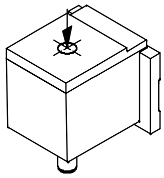
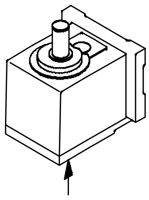
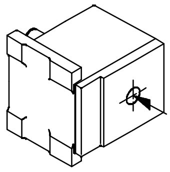
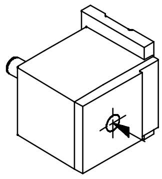
Tab. 8 Lubricant quantity in litre

Note:
➔ = Drive
A, M, N, W = Input design
90.....132 = Motor frame size
1A.....2K = Drive size

Maintenance

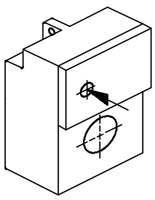
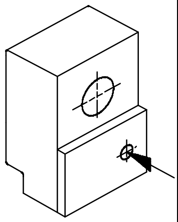
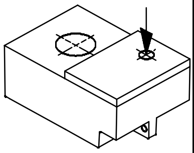
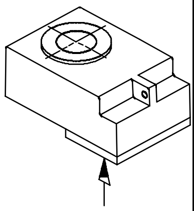
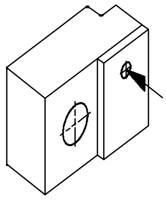
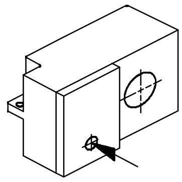
Maintenance operations

Replacing the lubricant

Lubricant quantity of helical gearboxes GST□□-2□; -3□ VC□						
Mounting position	**A** 	**B** 	**C** 	**D** 	**E** 	**F** 
GST03	0.17	0.17	0.17	0.17	0.17	0.17
GST04	0.35	0.45	0.3	0.6	0.3	0.3
GST05	0.55	0.7	0.6	0.95	0.45	0.45
GST06	1.1	1.3	1.1	1.7	0.9	0.9
GST07	1.8 2.2 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.4 2.8 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.2	2.7 3.4 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.5 1.9 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.5 1.9 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G
GST09	3.5 4.1 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	4.5 5.0 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	4.4	5.5 6.7 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.8 3.4 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.8 3.4 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G
GST11	7.5	8.5	8.0	10.5	6	6
GST14	13	16	14	18	9.5	9.5
Lubricant quantity of pre-stage GST□□-3□						
GST05	0.12 0.15 W ≥ 1C	0.3 0.35 W ≥ 1C	0.15	0.3 0.35 N ≥ 1B 0.4 W ≥ 1C	0.2	0.2
GST06	0.15	0.4	0.35	0.3 0.4 W ≥ 1C	0.3	0.3
GST07	0.3	0.7	0.5	0.55 0.65 M ≥ 90 A ≥ 80 N ≥ 1D	0.4	0.4
GST09	0.6	1.4	1.1	1.2	0.8	0.8
GST11	1.5 2.0 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.5 2.9 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.1	1.7 2.4 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.7	1.7
GST14	2.7 4.0 M ≥ 132 A ≥ 112 N ≥ 1G	4.6 5.2 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	4.3	3.2 4.1 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	3	3

Tab. 9 Lubricant quantity in litre

Note:
- ➔ = Drive
- A, M, N, W = Input design
- 90.....132 = Motor frame size
- 1A.....2K = Drive size

Lubricant quantity of low-profile gearboxes GFL□□-2; -3						
Mounting position	A 	B 	C 	D 	E 	F 
GFL04	0.65	0.6	1.05	0.8	0.75	0.35
GFL05	1.3	1.4	1.7	1.7	1.3	0.7
GFL06	2.0	2.1	3.0	2.8	2.1	1.2
GFL07	4.3	3.8	6.3	5.8	4.6	2.5
GFL09	8.9	7.6	13	11.3	9.5	5.2
GFL11	16	15	25	21	20	9.0
GFL14	32	36	47	42	27	28
Lubricant quantity of pre-stage GFL□□-3						
GFL05	0.2	0.2	0.15	0.3 0.35 N ≥ 1B 0.4 W ≥ 1C	0.12 0.15 W ≥ 1C	0.3 0.35 W ≥ 1C
GFL06	0.3	0.3	0.35	0.3 0.4 W ≥ 1C	0.15	0.4
GFL07	0.4	0.4	0.5	0.55 0.65 M ≥ 90 A ≥ 80 N ≥ 1D	0.3	0.7
GFL09	0.8	0.8	1.1	1.2	0.6	1.4
GFL11	1.7	1.7	2.1	1.7 2.4 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.5 2.0 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.5 2.9 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G
GFL14	3.0	3.0	4.3	3.2 4.1 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.7 4.0 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	4.6 5.2 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G

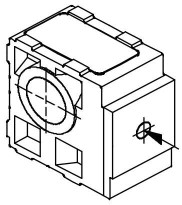
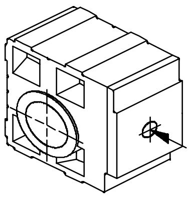
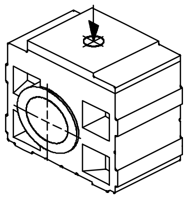
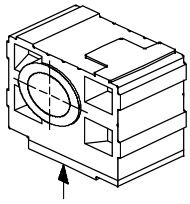
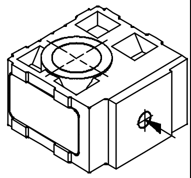
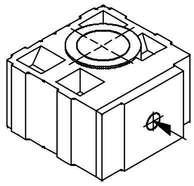
Tab. 10 Lubricant quantity in litre

Note:
- ➔ = Drive
- A, M, N, W = Input design
- 90.....132 = Motor frame size
- 1A.....2K = Drive size

Maintenance

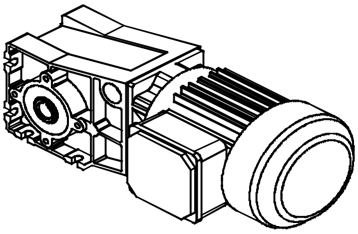
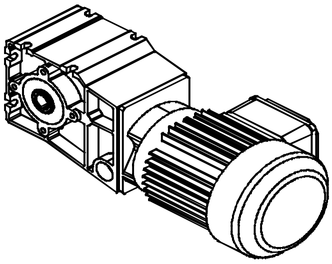
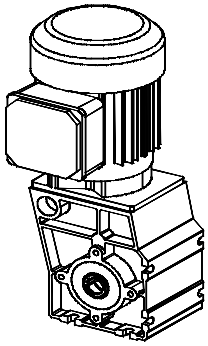
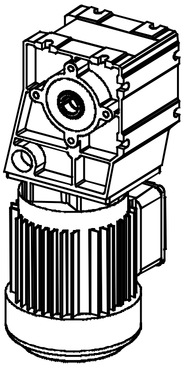
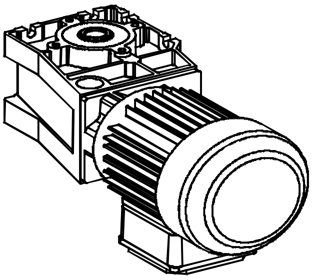
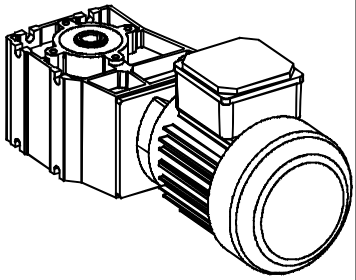
Maintenance operations

Replacing the lubricant

Lubricant quantity of right-angle gearboxes						
Mounting position	**A** 	**B** 	**C** 	**D** 	**E** 	**F** 
Helical bevel gearboxes GKS□□-3; -4						
GKS04	0.8	1.4	1.5	1.1	1.3	1.3
GKS05	1.4	2	2.1	1.7	1.9	1.9
GKS06	2.4	3.6	4	3	3.6	3.6
GKS07	4.5	6.7	7.7	5.6	6.5	6.5
GKS09	6	14	16	10	14	14
GKS11	11.5	27	29	21	25	25
GKS14	21	50	56	38	47	47
Helical-worm gearboxes GSS□□-2; -3						
GSS04	0.5	1.0	1.0	1.0	0.8	0.8
GSS05	1.2	1.7	1.7	1.7	1.4	1.4
GSS06	1.8	3.0	3.0	3.0	2.6	2.6
GSS07	3.6	5.6	5.9	5.6	4.8	4.8
Lubricant quantity of pre-stage GKS□□-4 / GSS□□-3						
G□□05-□	0.12 0.15 W ≥ 1C	0.3 0.35 W ≥ 1C	0.15	0.3 0.35 N ≥ 1B 0.4 W ≥ 1C	0.2	0.2
G□□06-□	0.15	0.4	0.35	0.3 0.4 W ≥ 1C	0.3	0.3
G□□07-□	0.3	0.7	0.5	0.55 0.65 M ≥ 90 A ≥ 80 N ≥ 1D	0.4	0.4
G□□09-□	0.6	1.4	1.1	1.2	0.8	0.8
G□□11-□	1.5 2.0 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.5 2.9 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	2.1	1.7 2.4 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	1.7	1.7
G□□14-□	2.7 4.0 M ≥ 132 A ≥ 112 N ≥ 1G	4.6 5.2 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	4.3	3.2 4.1 M ≥ 132 A ≥ 112 N ≥ 1G W ≥ 1G	3	3

Tab. 11 Lubricant quantity in litre

Note:
- ➔ = Drive
- A, M, N, W = Input design
- 90.....132 = Motor frame size
- 1A.....2K = Drive size

Lubricant quantities - Bevel gearboxes GKR			
Mounting position	**A** 	**B** 	**C** 
GKR03	0.35	0.35	0.4
GKR04	0.4	0.5	0.7 / 0.8 *
GKR05	0.8	1.3	1.5 / 1.6 *
GKR06	1.5	2.3	3.0 / 3.2 *
Mounting position	**D** 	**E** 	**F** 
GKR03	0.35	0.35	0.35
GKR04	0.7	0.6	0.4
GKR05	1.4	1.5	1.0
GKR06	2.6	3.0	1.8

Tab. 12 Lubricant quantity in litre
* for output design V□K

6.3 Repair

Lenze recommends that repairs are carried out by the Lenze Service.

6.4 Spare-parts list for geared motors

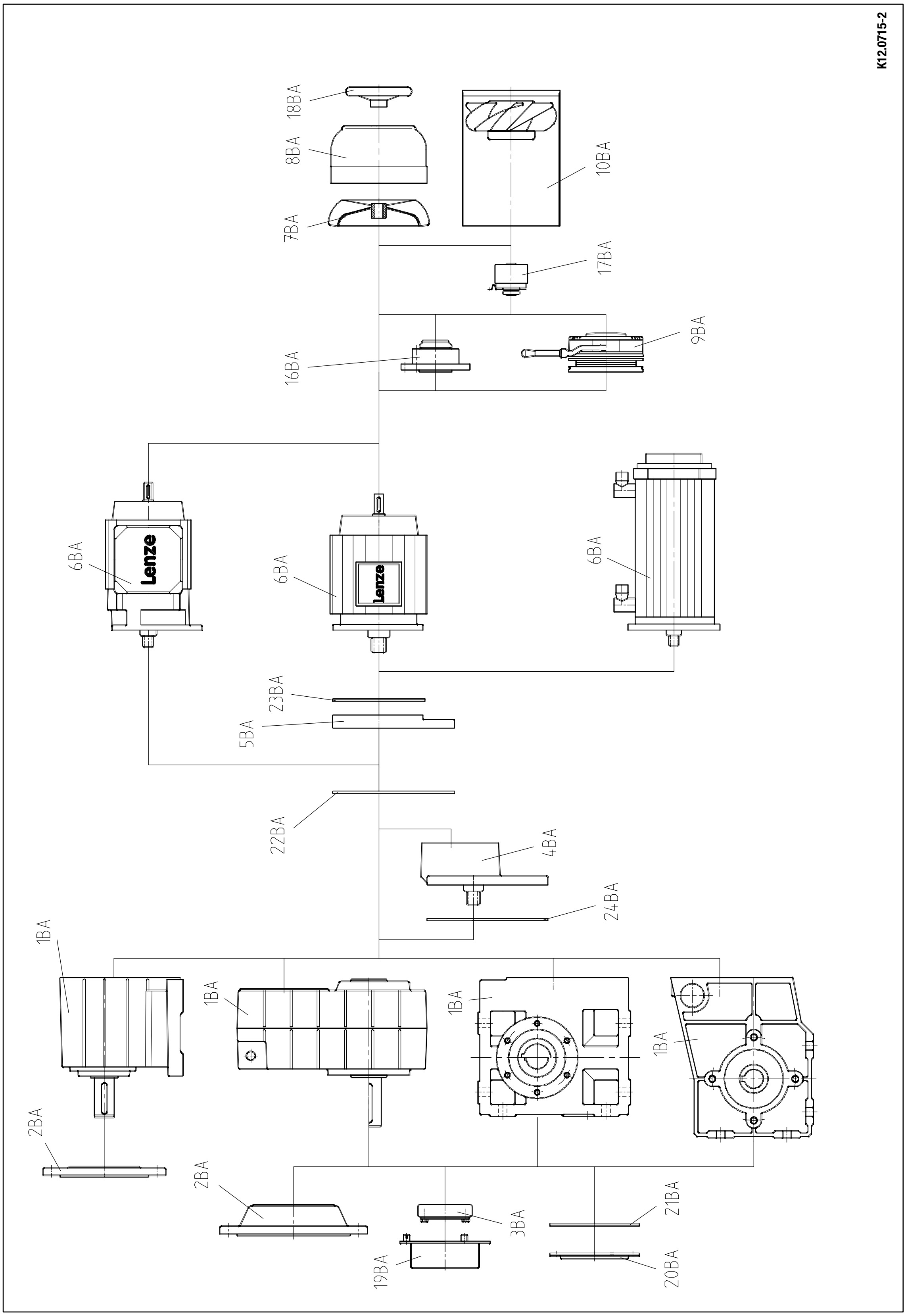


Fig. 8

6.5 Spare-parts list for gearboxes and gearboxes with variable speed drives

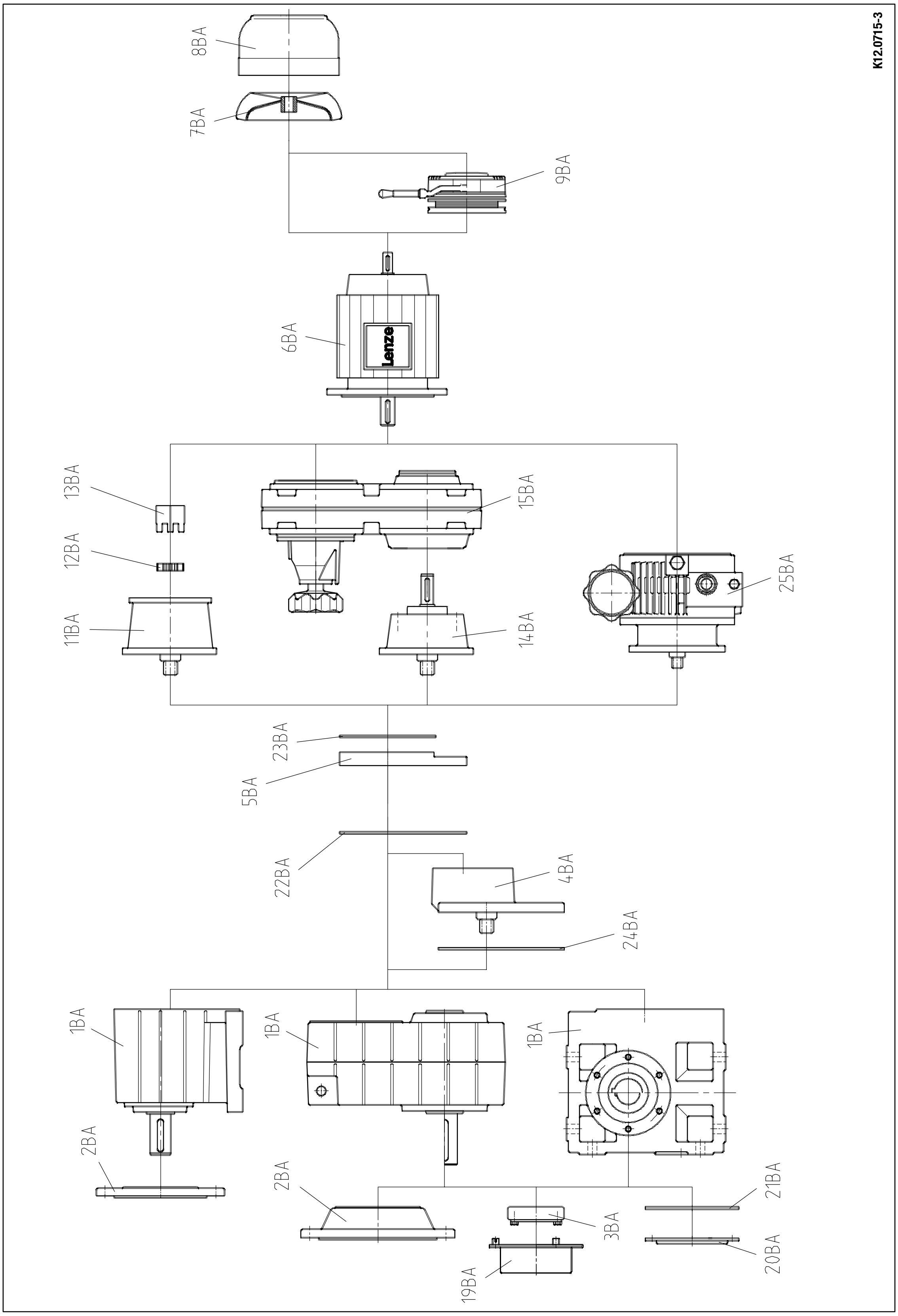


Fig. 9

6.6 Order form

Recipient: Lenze

Postal code/City: ____________________

Fax-no.: ____________________

all G☐☐

Sender

Company ____________________ Customer no. ____________________

Street / P.O. Box ____________________ Order no. ____________________

Postal code/City ____________________ Issued by ____________________

Delivery address ____________________ Phone ____________________

____________________ Fax ____________________

Invoice recipient* ____________________ Date of delivery ____________________

Date ____________ Signature ____________________

* Please indicate if other than sender

LENZE type number: ____________

Order number: ____________

Item	Name	Pieces
1BA	Basic gearbox GST ☐☐	
	Basic gearbox GFL ☐☐	
	Basic gearbox GKS ☐☐	
	Basic gearbox GKR ☐☐	
	Basic gearbox GSS ☐☐	
2BA	Output flange	
3BA	Shrink disk	
4BA	Prestage	
5BA	Intermediary cover	
6BA	Motor	
7BA	Fan	
8BA	Fan cover	
9BA	Spring-appplied brake	
10BA	Separate fan	
11BA	Flange	
12BA	Toothed ring	
13BA	Clutch hub	
14BA	Free input shaft	
15BA	Variable-speed drive	
16BA	Backstop	
17BA	Speed / position encoder	
18BA	Hand wheel	
19BA	Shrink disk cover	
20BA	Hollow shaft cover	
21BA	Seal	
22BA	Seal	
23BA	Seal	
24BA	Seal	
25BA	Variable speed planetary drive	

Fold-out page - spare-parts lists

7 Troubleshooting and fault elimination

If any malfunctions should occur during operation of the drive system, please check the possible causes using the following table. If the fault cannot be eliminated by one of the listed measures, please contact the Lenze Service.

Fault	**Possible cause**	**Remedy**
Drive does not start	Voltage supply interrupted	Check connection
	Wrong electrical connection	Check if nameplate matches voltage supply
	Excessive load	Reduce load Check drive - machine combination
Motor running, gearbox at standstill	Fixing elements are missing or defective	Check attachment
	Gearbox is defective	Contact Lenze Service
Unusual noise	Overload	Reduce load Check drive - machine combination
	Damage in the gearbox or motor	Contact Lenze Service
Excessive temperature	Overload	Reduce load Check drive - machine combination
	Insufficient heat dissipation	Improve cooling air supply Clean gearbox/motor
	Insufficient lubricant	Refill lubricant as specified
Loose fixing elements	Vibrations	Avoid vibrations

8 Waste disposal

Protect the environment! Packing material can be recycled.

What?			**Where?**
Transport material	Pallets		Return to the manufacturer or forwarder
	Packaging material		Cardboard boxes to waste paper Plastic to plastic recycling or waste material Reuse or dispose of wood shaving
Lubricants	Oil, grease		Dispose according to current regulations
Components	Housing: Bearings, shafts, gear wheels: Seals	grey cast iron steel special waste	Separate valuable substances and dispose

Manufacturer's declaration

Lenze

Gearboxes

Lenze Drive Systems GmbH
Postfach 10 13 52
D-31763 Hameln

Site: Bösingfeld
Breslauer Straße 3
D-32699 Extertal
Phone (05154) 82-0
Fax (05154) 82-15 75

Manufacturer's Certification

We herewith certify that the below listed products are intended for assembly into a machine or for assembly with other elements to form a machine. Commissioning of the machine is prohibited before it is proven that it corresponds to the EC regulation 98/37/EC.

Product:	**Type designation:**
Low-profile gearboxes and geared motors	GFL
Helical gearboxes and geared motors	GST, 12.6☐☐
Helical bevel gearboxes and geared motors	GKS, 12.5☐☐
Bevel gearboxes and geared motors	GKR
Helical worm gearboxes and geared motors	GSS, 52.1☐☐
Variable speed belt drives and geared motors	G☐☐-K 11.1☐☐, 11.2☐☐, 11.4☐☐
Variable speed drives and geared motors	G☐☐-D 11.7☐☐
Shaft-mounted gearboxes	12.4☐☐
Worm gearboxes and geared motors	52.3☐☐, 52.4☐☐, 52.5☐☐

Applied standards and regulations:

EN 292 part 1

EN 292 part 2

Hameln, July 01, 2003

Nitidem

(Dr.-Ing. Etienne Nitidem)
Head of Research and Development
Department for Electromechanics

Notes

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